DGPIR: Degradation-Guided Prompting for All-in-One Radiometric Restoration of Level-1 Optical Satellite Imagery

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***Abstract*—Commercial optical satellite constellations have greatly increased the volume and diversity of Level-1 (L1) optical products acquired across different sensors, orbit groups, and land-cover scenes. Although standard radiometric calibration and relative radiometric correction remove most systematic detector- or channel-level response differences, weak postcorrection residuals may remain because of time-varying on-orbit response, residual sensor-dependent variations, and complex scene radiance distributions. These residuals, including cross-track low-frequency color distortion, residual striping, and random noise, are often subtle, spatially nonuniform, sensor- and band-dependent, and tightly coupled with genuine land-cover textures, making reliable postcorrection restoration difficult without introducing over-correction, spectral bias, or texture degradation. To address this problem, we propose DGPIR, a degradation-guided prompt-based all-in-one framework for radiometric restoration of L1 optical satellite imagery. DGPIR uses the radiometric residual category identified by scene-level quality inspection, together with image-derived degradation statistics, to generate scale-specific dynamic prompts. These prompts are injected into a Transformer backbone through prompt-aware attention (PAA) and prompt-gated feed-forward (PGF) modulation, enabling a shared model to adapt its restoration behavior to physically different residual categories. A dual-branch inference strategy is further designed to match the spatial scale of L1 residuals: a global branch estimates full-swath low-frequency color distortion, while a patch-based branch handles stripe and noise residuals under practical memory constraints. Experiments on representative L1 restoration tasks show that DGPIR achieves consistent improvements over state-of-the-art all-in-one restoration baselines. For low-frequency color distortion, DGPIR improves PSNR by up to 0.60 dB and achieves strong cross-track radiometric consistency, with profile mean absolute error (Profile MAE) reduced to 0.0031 and profile correlation (Profile Corr.) reaching 0.8145 under the high-intensity setting. For destriping, DGPIR improves PSNR by 2.09 dB over the strongest all-in-one baseline in the mild-stripe regime and maintains the lowest RMSE under severe stripe degradation. It also shows stable denoising behavior, improves mixed-degradation restoration, and produces physically meaningful low-frequency correction profiles on additional real L1 scenes from different orbit groups and satellite platforms. The model maintains practical inference efficiency, requiring 95.84 GFLOPs and 424.18-MB peak GPU memory. The project repository is available at https://github.com/zhaxia191812/DGPIR**

***Index Terms*—All-in-one image restoration (AiOIR), low-frequency color distortion correction, prompt learning, remote sensing image restoration (RSIR).**

I. Introduction

DRIVEN by advances in micro–nano satellite manufacturing, cost-efficient payload design, and flexible constellation deployment, commercial remote sensing is shifting from single-satellite operation toward large-scale collaborative satellite constellations [1], [2], [3]. These systems play an increasingly important role in time-critical Earth-observation applications, such as precision agricultural monitoring [4], [5] and rapid disaster response [6], [7], [8]. To provide high revisit frequency and broad spatial coverage, representative commercial constellations, including Planet [9] and Jilin-1 [10], now often consist of tens to hundreds of compact optical satellites. While this constellation-based acquisition mode greatly improves data availability, it also increases the volume and heterogeneity of Level-1 (L1) optical products, placing higher demands on reliable and efficient ground processing.

Standard ground radiometric calibration and correction pipelines estimate and compensate dominant detector- or channel-level response differences using calibration coefficients, relative gain/bias parameters, and response-consistency models. These procedures ensure the radiometric quality of the majority of L1 products. In large-scale optical satellite constellations, however, residual radiometric nonuniformity may still remain in a subset of quality-flagged L1 products when the actual on-orbit response is affected by calibration uncertainty, response nonlinearity, temporal drift, or residual interdetector/interchannel response differences. At the product level, such residual nonuniformity may appear as cross-track low-frequency color distortion, residual striping, or weak random noise.

These postcorrection residuals are different from severe radiometric degradation in raw sensor data and from generic corruptions commonly studied in natural-image restoration. They are usually subtle, spatially nonuniform, sensor- and band-dependent, and may be tightly coupled with genuine land-cover textures or large-scale scene radiance variations. For example, weak residual stripes can be visually amplified over water, desert, snow, or homogeneous farmland, but may be confused with repetitive urban structures, mountain

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shadows, or agricultural textures in complex scenes. Low-frequency color distortion may appear as a slowly varying cross-track gain bias over a full swath, and the affected bands can differ across cameras, orbits, or satellite platforms. These characteristics make reliable restoration difficult: the model should suppress the target residual while avoiding over-correction, spectral bias, and degradation of real scene structures.

In L1 optical satellite products after standard radiometric calibration and relative radiometric correction, residual radiometric nonuniformity is often weak, scene-dependent, band-dependent, and physically heterogeneous. It may appear as a full-swath cross-track gain bias, column-dependent residual striping, local random noise, or their coexistence. Calibration-oriented correction methods generally describe detector or channel response differences using calibration coefficients, relative gain/bias parameters, or response-consistency models [11], [12], [13]. Such models are effective for dominant and repeatable systematic errors, but weak postcorrection residuals may be influenced by time-varying on-orbit response and scene-dependent radiance distributions, making them difficult to fully characterize using fixed or simplified correction parameters. Existing destriping and denoising methods [14], [15], [16], [17], [18], [19], [20], [21] are also highly relevant, but many of them are developed under task-specific assumptions. Filtering, variational, and column-statistics-based methods usually assume that degradation components are separable from real scene radiance in the spatial, frequency, or statistical domain. This assumption becomes restrictive when low-frequency color distortion is mixed with genuine large-scale land-cover radiance variation, or when weak residual stripes resemble repetitive scene textures. Task-specific deep models further assume an isolated degradation category, whereas residual color bias, striping, and noise may coexist and interfere with each other in L1 products.

These characteristics lead to three methodological requirements for L1 residual radiometric restoration. First, the restoration process should be category-aware, because low-frequency color distortion, residual striping, and random noise correspond to different physical residual forms. Second, it should use image-dependent degradation statistics to adjust the correction strength under scene-specific residual patterns. Third, the inference context should match the physical scale of the residual: low-frequency color distortion requires full-swath cross-track information, whereas residual stripes and weak random noise are better handled with column-aware or local processing. Therefore, a unified framework for this problem should be understood as a degradation-conditioned residual modeling strategy: it should learn shared radiometric representations related to sensor response and scene statistics, while explicitly conditioning the restoration behavior on different residual categories and degradation states. All-in-one image restoration (AiOIR) methods [22], [23], [24], [25] provide a useful reference for organizing multiple restoration tasks within a shared framework [26]. However, most generic AiOIR models learn prompts or adaptive modulation from general visual degradation cues and are mainly optimized for natural-image corruptions. They do not explicitly model cross-track gain fields, column-dependent perturbations, or band-dependent radiometric response variations in push-broom L1 imagery.

To address these challenges, we propose DGPIR, a degradation-guided prompt-based all-in-one framework for radiometric restoration of L1 optical satellite imagery. In practical L1 production workflows, each scene is typically subjected to radiometric quality inspection by trained operators or automated quality-control algorithms, through which the dominant residual category, such as low-frequency color distortion, residual striping, or weak random noise, can be identified. DGPIR uses this quality-inspection-derived residual category, together with image-derived degradation statistics, to generate scale-specific prompt vectors. These prompts modulate both attention and feed-forward transformations in a Transformer backbone, allowing the shared model to adapt its feature representation to physically different residual categories. In addition, DGPIR adopts a dual-branch inference strategy for large-swath imagery: a global branch estimates full-swath low-frequency color distortion with complete cross-track context, while a patch-based branch handles local stripe and noise residuals under practical memory constraints. The main contributions of this article are summarized as follows.

1) We propose DGPIR, a task-conditioned all-in-one framework specifically designed for postcorrection radiometric residual restoration in L1 optical satellite products. Unlike generic prompt-based restoration models developed mainly for natural-image degradations, DGPIR explicitly incorporates the radiometric residual category identified by quality inspection and restores low-frequency color distortion, striping, and random noise within a shared weight space.
2) We propose a degradation-guided global prompt (DGGP) mechanism that converts quality-inspection-derived residual category and image-derived degradation statistics into dynamic scale-specific prompt vectors. These prompts provide explicit degradation guidance for the restoration backbone, enabling the network to modulate its feature representation according to different residual patterns and reducing feature confusion among physically heterogeneous tasks.
3) We design a dual-branch inference strategy that matches the spatial scale of different L1 residuals. The global branch estimates full-swath low-frequency color distortion with complete cross-track context, whereas the patch-based branch handles local stripe and noise residuals under practical memory constraints. The framework is further evaluated under mixed-degradation and cross-orbit/cross-sensor scenarios to examine the stability, physical consistency, and applicability of prompt-guided restoration in heterogeneous L1 imagery.

II. Related Works

This section reviews the current state of research regarding radiometric restoration for remote sensing imagery. We first survey task-specific restoration models tailored to specific

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degradation types, which constitute the prevailing paradigm in current research. Subsequently, we introduce "all-in-one" restoration frameworks and discuss their limitations when transferred to remote sensing scenarios.

A. Task-Specific Restoration Models

Based on operational radiometric quality-inspection feedback, this study focuses on quality-flagged L1 optical satellite images that still require postcorrection refinement. The residual categories considered here include cross-track low-frequency color distortion, residual striping, and weak random noise.

Low-frequency radiometric nonuniformity and cross-track color inconsistency have been widely studied in relative radiometric calibration and correction for wide-swath optical satellite cameras. For sensors composed of multiple CCDs, focal-plane modules, or wide-field-of-view optical assemblies, existing studies have modeled response inconsistency caused by factors such as detector gain differences, vignetting, chromatic aberration, temperature-dependent response variation, and temporal drift [11], [12], [13]. Representative approaches include temperature-normalized relative radiometric correction, vignetting and drift correction for wide-field cameras, and relative correction models for multi-CCD systems with chromatic aberration. These studies provide important physical and statistical foundations for improving radiometric uniformity in standard satellite production pipelines.

Different from these calibration-oriented methods, this work focuses on the postcorrection refinement of quality-flagged L1 products. The residual fields considered here are usually weak and may vary across bands, cameras, or acquisition conditions, while being coupled with genuine large-scale land-cover radiance variations. Therefore, the restoration model should suppress radiometric residuals without altering real scene-dependent radiometric structures. Traditional image-processing methods, such as spatial-domain filtering [27], [28], [29], transform-domain filtering [30], [31], [32], [33], and variational methods [34], [35], have been used for remote sensing image restoration (RSIR). These methods are often interpretable and can incorporate explicit physical or statistical assumptions, but their shallow representations and idealized degradation models limit their ability to decouple weak radiometric residuals from complex scene content.

With the development of deep learning, many end-to-end models have been proposed for specific restoration tasks. For destriping, CNN-based methods such as DLS-NUC [36] and HSI-DeNet [37] learn mappings between stripe-contaminated observations and clean images by exploiting spatial and spectral dependencies. TSWEU [38] introduces wavelet-domain modeling into a dual-stream CNN framework to enhance directional-frequency representation, while DestripeCycleGAN [15] reduces the reliance on paired clean references by learning stripe distributions in an unsupervised manner. These methods have shown strong capability in suppressing structured stripe patterns when the stripe pattern is consistent with the assumed training or modeling condition.

For denoising, DnCNN [20] models the residual noise component through deep convolutional residual learning, and BRDNet [19] improves CNN-based denoising by introducing batch renormalization for more stable training and noise suppression. In remote sensing and hyperspectral imagery, SSCDN [39] explicitly exploits spatial–spectral correlations to preserve structural and spectral consistency, while HCANet [40] combines convolutional and attention mechanisms to enhance local and global feature representation. These denoising models are effective for random or signal-dependent noise, but they are usually optimized for a single degradation type and may not directly address structured radiometric residuals such as striping or low-frequency color distortion.

Overall, task-specific restoration methods remain effective under their assumed degradation models, especially when a single degradation component is isolated and its spatial or statistical structure is stable. Their limitation in L1 residual radiometric restoration lies in the isolated-degradation assumption. In coupled L1 residuals, different residual components do not simply add to the image independently: low-frequency gain bias changes the global radiometric baseline and band statistics, residual striping changes columnwise profile cues, and random noise perturbs local statistics used by stripe and profile estimation. When these components coexist, correcting one component independently may alter the evidence used to estimate another, leading to under-correction, over-correction, or texture attenuation. This coupling indicates that a set of independently optimized single-task models is not equivalent to a degradation-conditioned residual restoration model. Therefore, DGPIR is motivated as a task-conditioned framework that learns shared radiometric representations while using degradation guidance to separate physically different residual modes and condition the inverse mapping for each residual category.

B. All-in-One Restoration Models

Multitask restoration frameworks have been investigated to model multiple degradation modes within a shared conditional restoration space. In scenarios where the degradation type is fixed and isolated, task-specific models remain effective because their architectures and optimization objectives can be closely matched to the target degradation. By contrast, AiOIR is more relevant when different degradation modes may appear separately or jointly, and when the restoration model needs to learn degradation-dependent inverse mappings rather than independent degradation-specific mappings.

Early unified restoration methods often relied on task-specific components or feature-space alignment. For example, Restormer [41] and SwinIR [42] achieve strong performance in image restoration, but they are commonly trained separately for different tasks. IPT [43] retains task-specific priors through multihead and multitail designs, while AirNet [22] employs contrastive learning to discriminate degradation representations in the latent space for blind restoration. More recent methods introduce adaptive capacity allocation or expert routing. LoRA-IR [44] uses degradation-aware routing to activate low-rank adaptation experts, and MoCE-IR [45] allocates restoration capacity according to degradation complexity. OneRestore [46] and AllRestorer [47] further exploit

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TABLE I
METHODOLOGICAL COMPARISON BETWEEN DGPIR AND REPRESENTATIVE ADAPTIVE AIOIR METHODS

Method	Prompt / condition source	Condition / prompt generation	Network adaptation strategy	Inference mechanism	Target scenario
PromptIR	Image-derived degradation representation	Learns degradation-aware prompts from input features	Injects learned prompts into restoration blocks for all-in-one restoration	Single general restoration path	Natural-image restoration
AdaIR	Adaptive image/frequency representation	Generates adaptive modulation according to degradation-related frequency information	Frequency-aware feature modulation	Single adaptive restoration path	Natural-image restoration
MoCEIR	Degradation complexity / expert routing	Uses complexity-aware routing or expert allocation	Mixture-of-experts style capacity adaptation	Expert-routed restoration path	Complex natural-image degradation
DGPIR	Quality-inspection-derived radiometric residual category + image-derived degradation statistics	Fuses residual category embedding and global degradation statistics into scale-specific prompts	Prompt-aware attention and prompt-gated feed-forward modulation with stage-wise prompt sharing	Task-conditioned dual-branch inference: global branch for low-frequency color distortion and patch branch for striping/noise	L1 optical satellite radiometric restoration

cross-attention or multimodal information to improve restoration under composite degradation scenarios. These methods demonstrate the feasibility of conditional or adaptive restoration within a shared network, but many of them are still designed around generic natural-image degradations.

Prompt-based all-in-one restoration provides a lightweight way to condition restoration behavior without introducing large task-specific branches. InstructIR [48] uses human-readable instructions to guide restoration, while PromptIR [23] learns degradation-aware prompts from input features and injects them into the restoration network. AdaIR [24] introduces frequency-aware adaptive modulation to handle diverse image corruptions. DFPIR [49] enhances degradation-aware feature perturbation, and HAIR [50] uses hypernetwork-based adaptation to improve the flexibility of all-in-one restoration. These methods demonstrate that prompt or adaptive modulation can effectively guide a shared restoration backbone.

To clarify these differences, Table I summarizes representative prompt-based all-in-one restoration methods from the perspectives of prompt source, prompt generation, network adaptation, inference mechanism, and target scenario.

However, directly transferring these generic AiOIR designs to postcorrection L1 optical satellite imagery remains challenging. The residuals considered in this work differ substantially in physical scale and radiometric manifestation: low-frequency color distortion is expressed as a global cross-track radiometric field, residual striping appears as a column-dependent structured perturbation, and random noise is mainly local and stochastic. Treating these physically different residuals as a single generic degradation may lead to feature interference, insufficient correction of weak residuals, or over-smoothing of genuine land-cover structures.

Therefore, the central challenge is to model physically heterogeneous and potentially coupled L1 residuals within a shared conditional restoration space. This motivates a prompt-guided restoration design that uses quality-inspection-derived residual categories and image-derived degradation statistics to condition feature modulation, while adapting the inference strategy to the physical scale of each residual category.

III. METHOD

A. Unified Degradation Model and Problem Formulation

Consistent with the problem definition in the Introduction, this section formulates the degradation process for postcorrection residuals in L1 optical satellite imagery, rather than raw sensor-level degradation. Let $I_{clean} \in \mathbb{R}^{H\times W}$ be the theoretically clean L1 image after standard radiometric calibration and correction, and let I_{low} be the observed L1 image affected by residual radiometric nonuniformity. The degradation process is formulated as

$$I_{low} = \mathcal{D}_k\left(I_{clean}\right) \tag{1}$$

where $\mathcal{D}_k(\cdot)$ denotes the kth type of postcorrection residual degradation. In this work, we consider three representative residual categories that are commonly observed in quality-flagged L1 optical products: low-frequency color distortion, residual striping, and weak random noise.

First, low-frequency color distortion is modeled as a cross-track multiplicative gain field. Wide-swath optical satellite sensors often rely on focal-plane assemblies composed of multiple detector modules or readout channels. Although detector response differences are compensated by ground calibration and relative radiometric correction, small residual inconsistencies may remain because of calibration uncertainty, response nonlinearity, temporal drift, or interchannel response variation. At the L1 product level, these residuals may appear as slowly varying cross-track color bias or local step-like radiometric discontinuities. Such distortions mainly affect the radiometric consistency of spectral bands, while the underlying spatial structures of the scene are expected to be preserved.

To simulate this residual for data-driven training, we construct a hybrid gain-generation strategy. Specifically, low-frequency color distortion is modeled as a multiplicative gain field $M \in \mathbb{R}^{1\times W}$ which consists of three forms:

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discrete stitching-type distortion, continuous gradient-type distortion, and mixed distortion. The first category is discrete stitching-type distortion. Assuming the sensor array consists of multiple detector segments, we partition the image width into nonoverlapping subregions. For each subregion, a bias value is randomly sampled from a uniform distribution $b_i \sim U(-1, 1)$. This component simulates step-like residual differences between detector modules or readout channels, denoted as $C_{step}(x)$.

The second category corresponds to continuous gradient-type color distortion. To simulate this phenomenon, we construct a gradient function $G_{grad}(x)$ using Gaussian filtering. We first generate a 1-D random noise vector and then apply a large-scale Gaussian smoothing convolution characterized by

$$I_{low} = \mathcal{D}_k\left(I_{clean}\right) \tag{2}$$

parameter $\sigma_{low-freq}$. This procedure yields a spatially continuous and smoothly varying waveform. Let $\tilde{c} = n * k_{\sigma_{low-freq}}$ be the raw convolution result. The final normalized gradient distortion c is obtained by

$$c = \frac{\tilde{c} - \mu_{\tilde{c}}}{|\tilde{c} - \mu_{\tilde{c}}|_\infty} \tag{3}$$

where $n \sim \mathcal{N}(0, 1)$ denotes a standard Gaussian noise sequence, and $*$ represents the 1-D convolution operation. $k_{\sigma_{low-freq}}$ refers to a 1-D Gaussian kernel with a standard deviation of $\sigma_{low-freq}$. $\mu_{\tilde{c}}$ represents the mean value of the sequence $\tilde{c}$, and $|\cdot|_\infty$ denotes the L_∞-norm, which scales the amplitude to the range $[-1, 1]$.

The simulated effects of continuous low-frequency color distortion with varying $\sigma_{low-freq}$ are illustrated in Fig. 1. The third category is mixed low-frequency color distortion. We employ a random weighting strategy to linearly superimpose the two aforementioned components

$$C_{\text{raw}} = \alpha \cdot C_{\text{step}} + (1 - \alpha) \cdot G_{\text{grad}} \tag{4}$$

where α regulates the mixing ratio between stitching-type discontinuities and smoothly varying gradient errors. The mixed curve is normalized and mapped to a predefined intensity scale, which controls the degradation strength and provides samples with different residual magnitudes. The final multiplicative gain field M is formulated as

$$M = \mathbf{1} + S \cdot \frac{C_{\text{raw}}}{|C_{\text{raw}}|_\infty}. \tag{5}$$

Consequently, the simulated image with low-frequency color distortion is obtained by

$$I_{\text{low-freq}} = I_{\text{clean}} \odot M \tag{6}$$

where $\mathbf{1}$ denotes a matrix of ones with the same dimensions as the image, and $\odot$ represents the elementwise multiplication. Examples of different simulated low-frequency color distortion types are illustrated in Fig. 2. This formulation provides a physically motivated degradation prior for learning cross-track radiometric correction. The multiplicative gain curve represents cross-track residual response variations after routine radiometric correction, while the step-like and continuous components correspond to intersegment discontinuities and

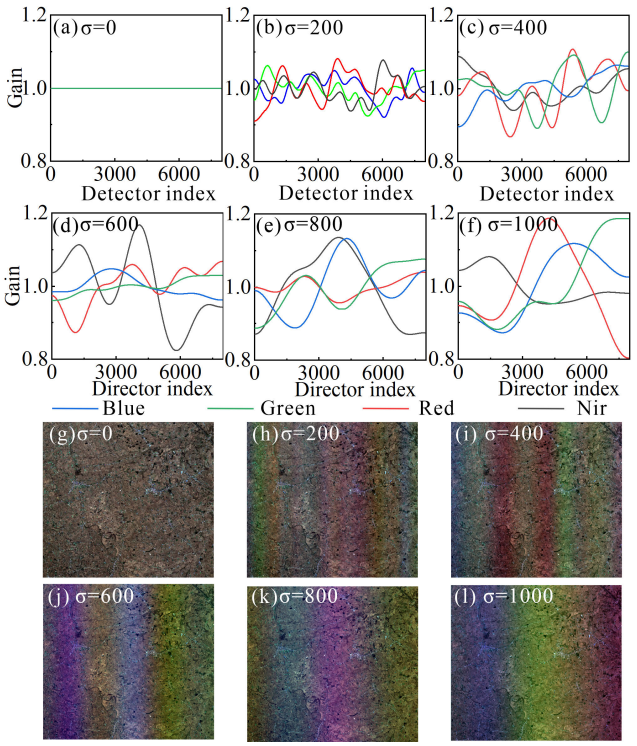


Fig. 1. Representative examples of simulated continuous low-frequency color distortion under different smoothing scales. (a)–(f) Multiplicative gain curves across the detector index under different values. (g)–(l) Corresponding simulated images. Blue, green, red, and black curves denote different spectral bands. Here, σ denotes the smoothing scale of the low-frequency gain curve, a larger σ produces a smoother gain curve and larger-scale spatial color distortion.

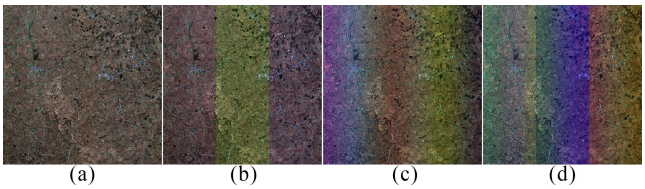


Fig. 2. Visual comparison of different simulated low-frequency color distortion types. (a) Clean L1 image. (b) Discrete stitching-type color distortion with interchip step-like residuals. (c) Continuous gradient-type color distortion with smooth spatial variations. (d) Mixed color distortion combining stitching discontinuities and continuous low-frequency gradients.

slowly varying detector/channel response drift, respectively. By varying the gain magnitude, smoothing scale, and mixture ratio during training, the network is exposed to a structured degradation space that reflects common low-frequency radiometric residuals in wide-swath push-broom imagery. Therefore, the restoration model learns the inverse mapping under physically constrained gain-field perturbations rather than under arbitrary color shifts.

Second, residual radiometric striping is modeled as a column-dependent radiometric perturbation involving both gain and bias terms. In push-broom optical imaging, pixels along the cross-track direction are associated with different detector elements or readout channels. After relative radiometric correction, small residual response differences may still remain because of imperfect nonuniformity correction,

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response nonlinearity, electronic noise, or interchannel gain drift. These residuals can appear as weak residual stripes extending along the along-track direction.

Based on this imaging characteristic, residual stripes are formulated as column-dependent perturbations that remain approximately consistent along the push-broom direction. For an image containing stripes, denoted as $I_{stripes}$

$$I_{stripes_{:,j}} = g_j \cdot I_{clean_{:,j}} + o_j + N_{:,j} \tag{7}$$

where g_j and o_j represent the multiplicative gain and additive bias factors for the jth column, respectively, and $N_{:,j}$ denotes random noise. To simulate realistic residual stripes while avoiding invalid pixel values, we model both g_j and o_j as independent truncated normal distributions

$$g_j \sim \mathcal{TN}\left(\mu_g, \sigma^2_{stripe_g}, [g_{min}, g_{max}]\right) \tag{8}$$

$$o_j \sim \mathcal{TN}\left(\mu_o, \sigma^2_{stripe_o}, [o_{min}, o_{max}]\right) \tag{9}$$

where $\mathcal{TN}(\mu, \sigma^2, [a, b])$ denotes a truncated normal distribution with mean μ, variance σ^2, and a truncation interval defined by $[a, b]$.

Third, weak residual noise is modeled as an additive radiometric perturbation after standard correction. Although calibration and correction pipelines normalize major gain and bias differences, small random components from photon noise, readout electronics, and correction uncertainty may remain in L1 products. Since this work focuses on weak postcorrection residuals, we approximate the remaining noise component as additive Gaussian noise with varying intensity. The generated noisy image is denoted as

$$I_{noisy} = I_{clean} + N\left(I_{clean}\right) \tag{10}$$

where $N(\cdot)$ represents the additive noise tensor. To simulate degradation scenarios under varying SNRs, the noise intensity σ_{noisy} is sampled from a uniform distribution

$$\sigma_{noisy} \sim \mathcal{U}\left(\sigma_{noisy_min}, \sigma_{noisy_max}\right). \tag{11}$$

For each pixel (i, j) in the image, the noise value follows a Gaussian distribution with mean 0 and standard deviation σ_{noisy}:

$$N_{i,j} \sim \mathcal{N}\left(0, \sigma^2_{noisy}\right). \tag{12}$$

B. Overall Architecture

As illustrated in Fig. 3, DGPIR adopts a hierarchical U-shaped encoder–decoder architecture and performs restoration under a task-conditioned setting. The task condition corresponds to the residual category identified by scene-level quality inspection, such as low-frequency color distortion, striping, or random noise. The DGGP module combines this residual category with image-derived degradation statistics to generate scale-specific prompts, which modulate feature transformations in the restoration backbone. In this way, DGPIR adapts to sensor- and condition-dependent residual variations through degradation-aware prompt modulation.

The three residual categories are associated with different spatial characteristics and therefore require different restoration contexts. Low-frequency color distortion is a full-swath radiometric residual and requires global cross-track context. Residual striping is a column-dependent structured perturbation, for which long-range spatial dependency modeling is important. Weak random noise is mainly local and stochastic, and can be restored under a patch-based processing strategy. Therefore, each residual category is assigned a distinct task condition, while restoration is performed by a shared backbone under degradation-specific prompt modulation.

To balance stability and inference efficiency, DGPIR adopts a stagewise prompt sharing strategy. The prompt vector is computed once at the entrance of each scale and then shared by all prompt modulation Transformer blocks (PMTBs) within the corresponding stage. This design provides consistent degradation guidance within each scale and avoids repeatedly estimating prompts from partially restored intermediate features.

The task-conditioned design addresses the heterogeneity and coupling of L1 residuals at three levels. At the degradation-prior level, DGGP introduces the radiometric residual category as an explicit condition and combines it with image-derived degradation statistics, so that the model receives both category-level residual guidance and scene-adaptive residual descriptors. At the feature-modulation level, the generated prompts are injected into both prompt-aware attention (PAA) and prompt-gated feed-forward (PGF) modules, enabling the backbone to adjust spatial dependency modeling and nonlinear feature transformation according to the dominant residual pattern. At the inference level, DGPIR separates global low-frequency correction from local stripe/noise restoration through a dual-branch strategy, so that residuals with different physical scales are processed with appropriate spatial context.

C. DGGP Generation

The DGGP module converts two complementary degradation priors into compact scale-specific prompt vectors. The first prior is the radiometric residual category obtained from scene-level quality inspection, which specifies the current restoration task, such as low-frequency color distortion, striping, or random noise. The second prior is an image-derived degradation descriptor extracted from the input feature map, which captures scene-dependent residual intensity and spatial distribution through global feature aggregation. As illustrated in Fig. 3, DGGP adopts a dual-stream formulation that fuses the explicit residual-category prior and the implicit image-derived degradation descriptor into a unified prompt representation.

For each scale s, the intent stream maps the task ID t to a learnable embedding

$$\boldsymbol{e_t} = \mathrm{Emb}\left(t\right). \tag{13}$$

In parallel, the statistics stream extracts a content-adaptive degradation descriptor from the feature map F_s. Global average pooling (GAP) is applied to aggregate spatial information into a channelwise descriptor

$$\boldsymbol{z_s} = \mathrm{GAP}\left(\boldsymbol{F_s}\right). \tag{14}$$

The descriptor is then refined by a lightweight bottleneck to obtain a compact degradation vector

$$\boldsymbol{d_s} = \phi_s\left(\boldsymbol{z_s}\right) \tag{15}$$

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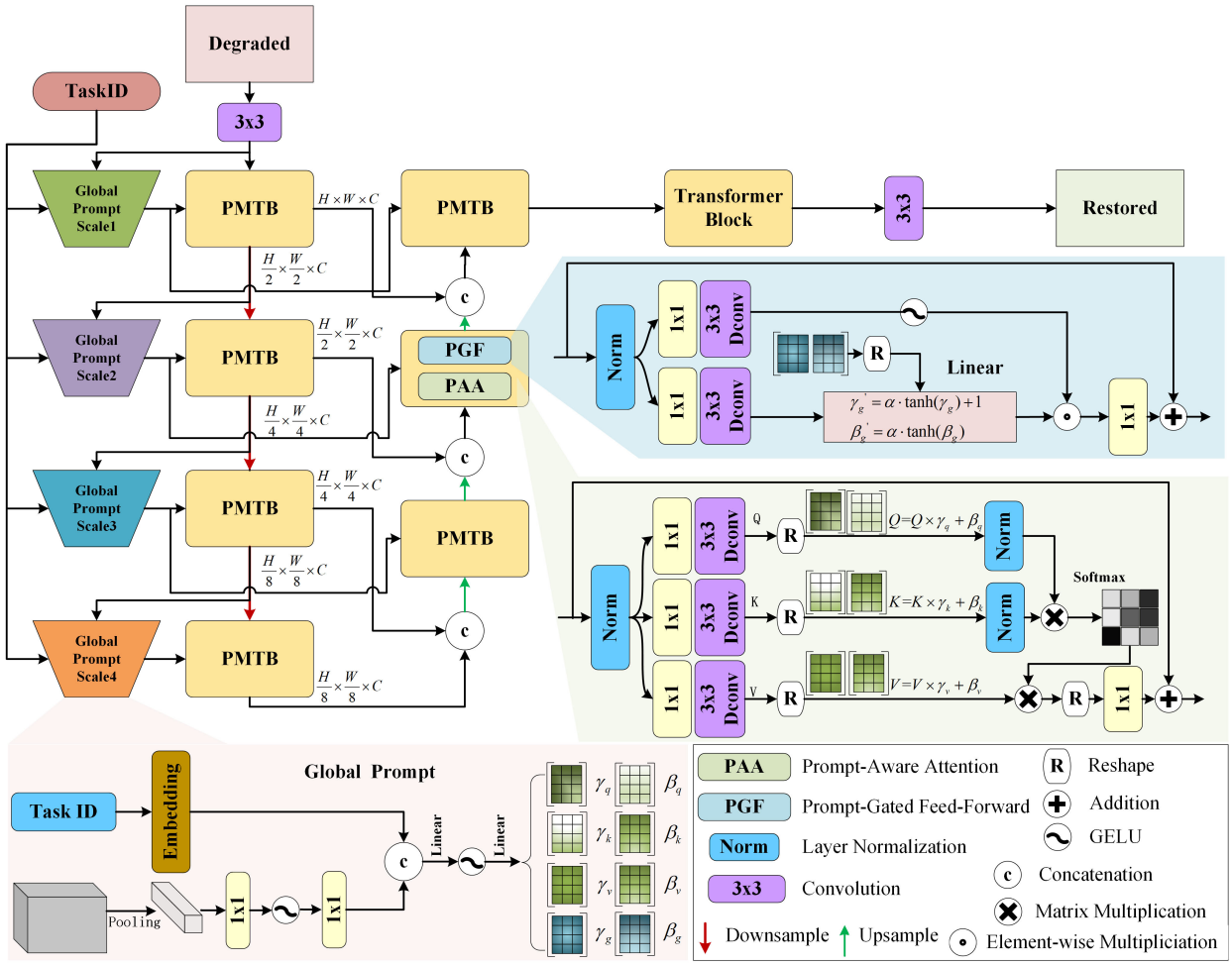


Fig. 3. Overview of the proposed DGPIR framework. DGPIR adopts a hierarchical U-shaped encoder–decoder architecture composed of PMTBs, followed by an additional Transformer block for feature refinement. The DGGP module jointly encodes the quality-inspection-derived residual-category code and global degradation features to generate scale-specific prompts, which are injected into the PAA and PGF modules within each PMTB for degradation-adaptive multiscale restoration.

$$\phi_s(\cdot) = \boldsymbol{W}_2^{(s)} \sigma\left(\boldsymbol{W}_1^{(s)}(\cdot)\right) \tag{16}$$

where $\sigma(\cdot)$ denotes a nonlinear activation and $\boldsymbol{W}_1^{(s)}$ and $\boldsymbol{W}_2^{(s)}$ are scale-specific learnable projections. The category embedding and degradation vector are concatenated to form a global prompt seed vector, which is then mapped by a small MLP to generate the final scale-specific prompt vector. The generated prompt provides the modulation parameters used by PAA and PGF, including the affine coefficients for attention projection modulation and the gating coefficients for feed-forward feature selection.

The prompt is computed once at the entrance of scale and shared by all PMTBs within the corresponding stage. Therefore, the final prompt used by each block is inherited from the scale-level category-statistics fusion result rather than independently re-estimated from intermediate restored features, ensuring consistent degradation guidance within each stage.

D. Prompt Modulation Transformer Blocks

A PMTB is introduced as the basic Transformer unit in DGPIR to enable task-conditioned feature transformation under heterogeneous and spatially nonstationary degradations. Given the input feature map $\boldsymbol{X}$ and the scale-specific prompt $\boldsymbol{p}_s$, PMTB follows a prenormalization residual design and sequentially applies PAA and PGF transformation

$$\boldsymbol{Y} = \boldsymbol{X} + \mathrm{PAA}\left(\mathrm{LN}(\boldsymbol{X}); \boldsymbol{p}_s\right) \tag{17}$$

$$\boldsymbol{Z} = \boldsymbol{Y} + \mathrm{PGF}\left(\mathrm{LN}(\boldsymbol{Y}); \boldsymbol{p}_s\right) \tag{18}$$

where $\mathrm{LN}(\cdot)$ denotes the layer normalization, and the scale-specific prompt $\boldsymbol{p}_s$ is generated by DGGP and shared within stage s. By injecting $\boldsymbol{p}_s$ into both the attention pathway and the feed-forward pathway, PMTB transforms the backbone from a fixed mapping into an input-adaptive conditional mapping, which helps the backbone adapt to degradation variations across scenes, sensors, and spectral bands.

PAA conditions the attention projections using affine modulation derived from the prompt. Let $\tilde{X} = \mathrm{LN}(X)$. A convolutional projection is used to obtain Q_0, K_0, and V_0, which are reshaped into h attention heads, where h is the number of heads. DGGP provides prompt-dependent coefficients $\{\gamma_q^{(s)}, \beta_q^{(s)}, \gamma_k^{(s)}, \beta_k^{(s)}, \gamma_v^{(s)}, \beta_v^{(s)}\}$, the prompt-aware projections are then obtained by FiLM-style affine transformation

$$\boldsymbol{Q} = \boldsymbol{Q_0} \odot \gamma_q^{(s)} + \beta_q^{(s)} \tag{19}$$

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$$\boldsymbol{K} = \boldsymbol{K_0} \odot \gamma_k^{(s)} + \beta_k^{(s)} \tag{20}$$

$$\boldsymbol{V} = \boldsymbol{V_0} \odot \gamma_v^{(s)} + \beta_v^{(s)}. \tag{21}$$

To stabilize attention computation, $\boldsymbol{Q}, \boldsymbol{K}$, and $\boldsymbol{V}$ are normalized along the head dimension, yielding $\bar{\boldsymbol{Q}}, \bar{\boldsymbol{K}}$, and $\bar{\boldsymbol{V}}$. The attention response is computed as

$$\boldsymbol{A} = \text{Softmax}\left(\tau \bar{\boldsymbol{Q}} \bar{\boldsymbol{K}}^{\top}\right), \text{PAA}\left(\tilde{X}; \boldsymbol{p_s}\right) = \text{Proj}\left(\boldsymbol{A}\bar{\boldsymbol{V}}\right) \tag{22}$$

where τ is a learnable temperature and Proj$(\cdot)$ denotes the output projection. This design allows the attention mechanism to adapt its query–key matching and value aggregation according to the global degradation state encoded in $\boldsymbol{p_s}$.

PGF conditions the gated branch in the feed-forward module to provide a degradation-aware nonlinear transformation. Given $\tilde{Y} = \text{LN}(Y)$, a pointwise projection followed by a depthwise convolution produces two hidden branches U_1 and U_2, where D is the hidden width. DGGP outputs prompt-dependent coefficients $\gamma_g^{(s)}$ and $\beta_g^{(s)}$ constrained as

$$\gamma_g^{(s)} = \alpha \tanh\left(\hat{\gamma}_g^{(s)}\right) + 1, \beta_g^{(s)} = \alpha \tanh \hat{\beta}_g^{(s)}. \tag{23}$$

The gated branch $\boldsymbol{U_2}$ is then modulated by the prompt vector $\boldsymbol{p_s}$ via an affine transformation

$$\boldsymbol{U_2'} = \boldsymbol{U_2} \odot \gamma_g^{(s)} + \beta_g^{(s)}. \tag{24}$$

The final output of the PGF is obtained through elementwise gating and an output projection

$$\text{PGF}\left(\tilde{Y}; \boldsymbol{p_s}\right) = \text{Proj}\left(\sigma\left(\boldsymbol{U_1}\right) \odot \boldsymbol{U_2'}\right) \tag{25}$$

where $\sigma(\cdot)$ denotes the GELU activation function and Proj$(\cdot)$ maps the hidden D representation back to the original channel dimension. By modulating the gate branch with p_s, PGF adaptively adjusts the effective nonlinearity and feature selectivity according to the specific degradation condition. This dynamic adjustment allows PGF to adjust restoration strength under different degradation conditions, helping mitigate under-restoration under severe degradations and over-smoothing under mild degradations.

IV. Experiments

We begin by introducing the datasets, experimental settings, and evaluation metrics used in this study. We then evaluate DGPIR on representative L1 radiometric restoration tasks with different degradation types and severity levels and compare it with both all-in-one restoration baselines and task-specific or traditional methods. In addition, we analyze mixed-degradation performance, cross-orbit and cross-sensor physical consistency, robustness, efficiency, and the contribution of key architectural components through ablation studies.

A. Datasets

We choose high-resolution imagery from the Jilin-1 KF02B satellite series as the main data source for this study. The KF02B series currently consists of seven satellites (KF02B01–KF02B07), providing a stable and scalable data supply for regional-scale Earth observation applications. Its push-broom optical payload supports ultrawide-swath observations (>150 km), which reduces mosaicking requirements and helps maintain radiometric consistency over regional extents. The system provides 0.5-m spatial resolution in the panchromatic (PAN) band and 2-m resolution in four multispectral (MS) bands (blue, green, red, and near-infrared), enabling joint analysis of fine spatial details and spectral information. Detailed specifications of the KF02B constellation are summarized in Table II.

TABLE II
Basic Specifications of Jilin-1 Sensors Used in This Study

Parameter	Specification	
Satellite Platform	Jilin-1 KF02B	Jilin-1 GF03D
Ground Resolution	PAN/MSS:0.5m/2m	PAN/MSS:0.75m/3m
Swath Width	> 150 km	> 17 km
Spectral Range	Band 0:PAN(436~712nm) Band 1:Blue(430~533nm) Band2:Green(491~597nm) Band3:Red(617~716nm) Band4: NIR(749~910nm)	Band 0:PAN(451~712nm) Band 1:Blue(431~520nm) Band2:Green(512~613nm) Band3:Red(607~698nm) Band4: NIR(764~903nm)
Quantization Bits	PAN: 14 bits; MSS: 16 bits	PAN: 12 bits; MSS: 12 bits

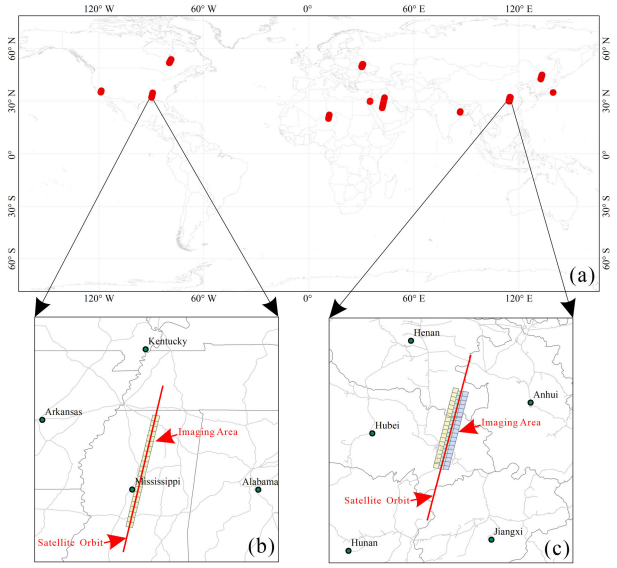
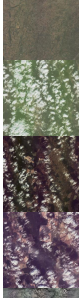


Fig. 4. Spatial distribution and representative orbit-imaging geometry of the additional real L1 validation data. (a) Global distribution of the selected validation orbit groups, where each red footprint denotes one orbit group rather than an individual image. (b) and (c) Enlarged examples of adjacent L1 imaging scenes acquired along representative satellite orbits. The red line indicates the satellite orbit direction, and the colored strips denote the adjacent imaging scenes or adjacent swaths. These data include single-satellite single-orbit cases, wide-swath observations, and adjacent PMS-camera imaging cases and are used to evaluate the applicability of DGPIR under different satellite platforms, orbit geometries, and geographic conditions.

To examine the applicability of DGPIR beyond the original KF02B training setting, we further collected additional operational L1 scenes from different satellite platforms, PMS cameras, orbit groups, and geographic regions. The additional validation data include KF02B05 scenes acquired by multiple PMS cameras under different orbit groups, as well as several scenes from the Jilin-1 GF03D series. Fig. 4 illustrates the spatial distribution and representative orbit-imaging geometry of these additional validation data. Table III further summarizes

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TABLE III
REAL L1 ORBIT GROUPS USED FOR CROSS-SENSOR APPLICABILITY ANALYSIS

Study area	Sensor	Coordinates	Imaging date	Thumbnail image
a	KF02B05 PMS06/PMS07/PMS08	N32.3°, E114.8°	2025/11/27	
b	KF02B05 PMS06/PMS07/PMS08	N51.1°, E30.6°	2025/12/13	
c	GF03D23	N53.8°, W8.8°	2026/05/14	
d	GF03D23	N45.0°, E133.2°	202/605/14	
e	GF03D23	N30.1°, E34.9°	2026/05/17	
f	GF03D23	N22.1°, E11.6°	2026/05/11	
g	GF03D48	N36.0°, W118.6°	2026/05/12	
h	GF03D48	N34.7°, W89.2°	20260513	
i	GF03D48	N32.0°, W43.2°	2026/05/14	
j	GF03D42	N16.3°, W16.1°	2025/12/22	
k	GF03D52	N27.1°, W4.6°	2025/12/22	

the corresponding satellite, camera, acquisition date, geographic location, and representative scene thumbnails. Since clean reference images are unavailable for these operational L1 products, these data are not used for paired PSNR/SSIM evaluation. Instead, they are used for cross-sensor applicability analysis based on the estimated low-frequency gain profiles. This setting is closer to practical L1 quality enhancement, where the main concern is whether the model can estimate physically meaningful cross-track correction fields under different sensors and imaging conditions.

To construct a high-quality ground-truth (GT) dataset, we selected 1000 high-quality L1 images from the KF02B satellite series, covering diverse land-cover types. These images were subjected to manual radiometric quality inspection to ensure the absence of significant radiometric residuals, such as striping, dead lines, or strong sensor noise. Scene diversity was also considered, including urban areas, road networks, farmlands, mountainous regions, and deserts.

Considering GPU memory constraints, we randomly cropped multiple patches from each original ultrawide swath image. This process yielded 64 874 patches in total, of which 63 874 were used for training, and 1000 were reserved for testing. Using these high-quality L1 patches as clean references, we simulate postcorrection radiometric residuals based on the degradation models described in Section III-A.

B. Experimental Settings

We categorize the restoration problem into three task conditions according to the radiometric residual category: $t = 0$ for low-frequency color distortion, $t = 1$ for destriping, and $t = 2$ for denoising. During training and testing, each sample is paired with the task index corresponding to its residual category, following the quality-inspection-feedback setting considered in this study.

For low-frequency color distortion, the number of sensor segments K is sampled from $[2, 4]$. The smoothing scale $\sigma_{\text{low-freq}}$ is sampled from $[10, 40]$, and the gain magnitude α is sampled from $[0.8, 1.2]$. The probabilities of generating step-type, continuous-type, and mixed low-frequency distortions are set to 0.2, 0.6, and 0.2, respectively. The intensity factor S is sampled from $[0.02, 0.2]$.

For residual striping, the gain standard deviation is set to $\sigma_{\text{stripe},g} = 0.05$. For residual noise, the noise intensity σ_{noisy} is sampled from $[10/65\,535, 150/65\,535]$, covering scenarios from subtle noise to relatively strong interference.

To evaluate the behavior of DGPIR under more realistic L1 conditions, we additionally construct mixed-degradation test samples by sequentially applying the above degradation models to clean KF02B images. Three mixed settings are considered: low-frequency color distortion plus noise (C + N), striping plus noise (S + N), and the coexistence of low- frequency color distortion, striping, and noise (C + S + N). The degradation parameters are sampled from the same ranges as those used in the single-degradation experiments to maintain consistency.

Our model is implemented using PyTorch and trained on four NVIDIA RTX 3090 GPUs. The Adam optimizer is used for parameter updates, with $\beta_1 = 0.9, \beta_2 = 0.999$, and weight decay of 1×10^{-4}. The objective function is the L_1 loss between the reconstructed image and the GT image. The initial learning rate is set to 3×10^{-4}. We adopt a cosine annealing with warm restarts learning-rate schedule. The minimum learning rates are set to 1×10^{-4} and 1×10^{-6}, respectively. The total batch size is set to 20, and the model is trained for 100k iterations.

DGPIR is instantiated as a four-level hierarchical encoder–decoder architecture. The number of Transformer blocks at each level is configured as [4, 6, 6, 8], with an additional refinement stage containing four blocks. The initial feature dimension is set to 48, and the number of attention heads progressively increases with network depth as [1, 2, 4, 8]. The feed-forward expansion factor is set to 2.66, and bias-free layer normalization is used.

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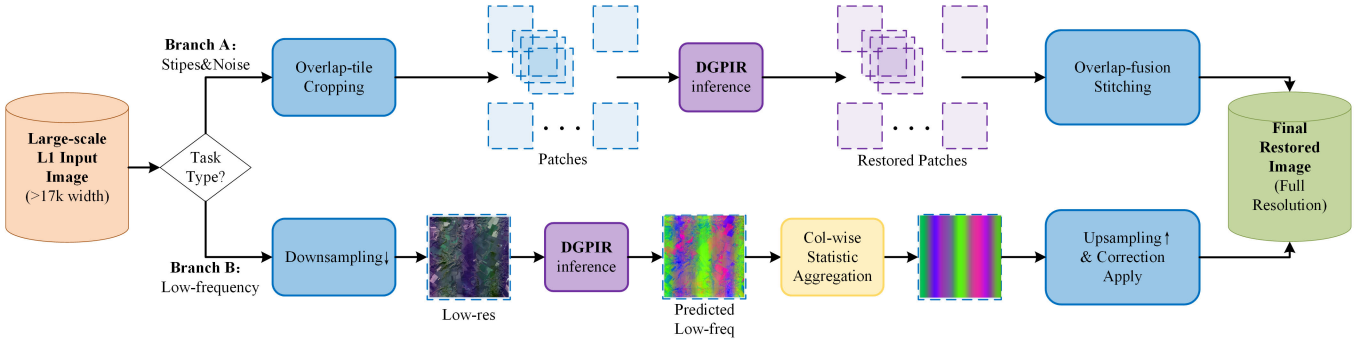


Fig. 5. Overview of the dual-branch inference pipeline of DGPIR for large-swath L1 image restoration. According to the quality-inspection-derived radiometric residual category, DGPIR selects the appropriate inference branch for each restoration pass. Branch A, used for residual striping and weak random noise (task ID = 1/2), performs full-resolution overlapping-tile cropping, patchwise DGPIR inference, and overlap-fusion stitching. Branch B, used for low-frequency color distortion (task ID = 0), performs low-resolution global inference to estimate the low-frequency correction field, followed by columnwise statistical aggregation, upsampling, and full-resolution correction application. For mixed-degradation inputs, the same network is applied sequentially with different task IDs from global low-frequency correction to local stripe/noise restoration.

Considering that L1 remote sensing imagery usually has ultralarge spatial resolution and that the physical scales of residual degradations differ across tasks, we design a task-conditioned dual-branch inference pipeline, as illustrated in Fig. 5. For local degradation tasks, including residual striping and weak random noise, we employ an overlapping patch-based strategy. The large-scale image is cropped into fixed-size patches for independent inference, and weighted fusion is applied in overlapping regions to reduce boundary discontinuities. In this branch, the shared DGPIR backbone is conditioned by the corresponding task prompt, enabling stripe removal and noise suppression with degradation-specific feature modulation while preserving full-resolution local details.

Low-frequency color distortion, in contrast, typically spans the entire image swath and exhibits slow spatial variation. Direct patch-based inference would discard global cross-track radiometric context and may lead to inconsistent correction between patches. Therefore, for low-frequency color distortion, the original large-scale L1 image is first bilinearly downsampled to a fixed inference size, enabling efficient processing while preserving the global radiometric trend over the full swath. The downsampled image is then fed into DGPIR, where the global prompt guides the network to predict a low-frequency distortion field at low resolution. Since low-frequency color distortion usually shows strong consistency along the push-broom direction, columnwise statistical aggregation is performed on the predicted field to extract a stable column response. A 1-D smoothing operation is further applied to suppress outliers and obtain a smooth columnwise correction curve. Finally, the resulting correction field is upsampled to the original resolution and applied to the full-resolution input image, thereby achieving global low-frequency correction while maintaining computational efficiency.

For mixed-degradation inputs, the quality-inspection process may return more than one radiometric residual category. In this case, DGPIR performs quality-inspection-guided sequential restoration using the same network weights but different task indices. The restoration order follows the physical scale of the residuals: low-frequency color distortion is first corrected using the global branch, followed by structured stripe removal and random noise suppression using the patch-based branch. This strategy avoids compressing heterogeneous residual categories into a single prompt and allows each restoration pass to be guided by the prompt corresponding to its dominant residual pattern.

To quantitatively evaluate the restoration performance, we adopt three widely used pixelwise metrics: PSNR [51], SSIM [52], and RMSE() [53]. For low-frequency color distortion, we further evaluate radiometric consistency using the spectral angle mapper (SAM) [54] and four task-specific metrics defined in Section IV-C, including normalized mean deviation (NMD), normalized standard deviation (NSD), profile mean absolute error (Profile MAE), and profile correlation (Profile Corr.).

C. Comparative Analysis of Restoration Performance

To comprehensively evaluate the restoration efficacy and practical applicability of DGPIR, we compare it with two groups of baselines. The first group includes five representative all-in-one restoration models, namely, PromptIR [23], AdaIR [24], MoCEIR [45], DFPIR [49], and HAIR [50], which are used to evaluate unified restoration capability across three typical categories of L1 radiometric residuals: low-frequency color distortion, residual striping, and weak random noise. The second group includes task-specific or traditional baselines for representative residual categories. For the low-frequency color-distortion task, traditional column-statistics-based correction methods, including CM, CMA, and CMM, are evaluated to assess the contribution of the proposed global correction branch under challenging cross-track radiance variations. For the destriping task, representative remote sensing destriping methods, including UTV [55], TSWEU [38], and WFAF [33], are adopted for direct comparison with algorithms explicitly designed for stripe removal. This section focuses on single-degradation restoration performance, while mixed-degradation cases are analyzed separately in Section IV-D.

For a fair comparison, all learning-based baselines were retrained on the same KF02B synthetic degradation dataset using the same training/testing split, degradation-generation

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TABLE IV
COMPARISON OF RESTORATION PERFORMANCE UNDER LOW-FREQUENCY COLOR DISTORTION AT DIFFERENT INTENSITY RANGES (PSNR/SSIM ↑ AND RMSE ↓). BEST RESULTS ARE SHOWN IN BOLD

range	PSNR↑			SSIM↑			RMSE↓		
degraded	53.5221	53.1584	52.9511	0.9987	0.9985	0.9982	139.3617	145.0319	148.4122
promptIR	54.5229	54.0706	53.6390	0.9992	0.9990	**0.9988**	126.7447	126.7447	138.6474
AdaIR	54.2371	53.8776	53.4943	0.9990	0.9988	0.9985	130.4458	135.2551	140.7476
MoCEIR	54.4141	53.9253	53.5149	0.9991	0.9989	0.9986	127.7389	134.2234	140.1469
DFPIR	54.4410	54.0038	53.5917	0.9991	0.9989	0.9986	127.6321	133.3280	139.1955
HAIR	54.2979	53.8948	53.4978	0.9608	0.9595	0.9538	129.7122	135.0672	140.7322
ours	**54.8967**	**54.2620**	**53.7939**	**0.9993**	**0.9991**	**0.9988**	**121.5711**	129.5617	**136.2304**

protocol, input patch size, optimizer type, training iterations, learning-rate schedule, loss function, and evaluation protocol. Only model-specific architectural settings and memory-constrained batch-size adjustments followed the corresponding implementations. Traditional nonlearning baselines were evaluated using their recommended parameter settings.

Quantitative results are first evaluated using three widely used pixelwise metrics: PSNR, SSIM, and RMSE, where higher PSNR/SSIM and lower RMSE indicate better reconstruction quality. For low-frequency color distortion, pixelwise metrics alone are not sufficient to characterize global radiometric consistency and cross-track profile preservation. Therefore, we additionally use SAM to evaluate spectral/color angular consistency, and define four task-specific radiometric metrics: NMD, NSD, Profile MAE, and Profile Corr.

For an image with spectral bands, NMD and NSD are defined as

$$\text{NMD} = \frac{1}{B}\sum_{b=1}^{B}\frac{|\mu_b(\hat{I}) - \mu_b(I)|}{\mu_b(I) + \epsilon} \tag{26}$$

$$\text{NSD} = \frac{1}{B}\sum_{b=1}^{B}\frac{|\sigma_b(\hat{I}) - \sigma_b(I)|}{\sigma_b(I) + \epsilon} \tag{27}$$

where I and $\hat{I}$ denote the reference and restored images, respectively, and μ_b and σ_b denote the mean and standard deviation of band b. and ϵ is a small constant used to avoid division by zero. NMD and NSD measure the relative deviations of bandwise radiometric mean and contrast, respectively.

To evaluate cross-track radiometric consistency, we average each image along the along-track direction and obtain a 1-D cross-track profile for each band

$$p_b(x) = \frac{1}{H}\sum_{y=1}^{H} I_b(y, x) \tag{28}$$

$$\hat{p}_b(x) = \frac{1}{H}\sum_{y=1}^{H} \hat{I}_b(y, x). \tag{29}$$

Profile MAE and Profile Corr. are then defined as

$$\text{Profile MAE} = \frac{1}{B}\sum_{b=1}^{B}\frac{1}{W}\sum_{x=1}^{W}|\hat{p}_b(x) - p_b(x)| \tag{30}$$

$$\text{Profile Corr.} = \frac{1}{B}\sum_{b=1}^{B}\rho(\hat{p}_b, p_b) \tag{31}$$

where $\rho(\cdot,\cdot)$ denotes the Pearson correlation coefficient. Lower NMD, NSD, SAM, and Profile MAE indicate better radiometric consistency, while higher Profile Corr. indicates better preservation of the cross-track radiometric profile.

Table IV presents the quantitative results of different all-in-one restoration models under three intensity ranges of low-frequency color distortion. DGPIR achieves the best or competitive performance across most pixelwise metrics. Specifically, its PSNR improves by 0.601, 0.377, and 0.276 dB over the second-best method under the three intensity ranges, respectively. The SSIM values are consistently high, all exceeding 0.999. In terms of RMSE, DGPIR achieves the lowest errors under the low- and medium-intensity settings, while remaining competitive under the high-intensity setting. These results indicate that the proposed model is effective in correcting low-frequency color distortion.

To further evaluate low-frequency radiometric fidelity beyond pixelwise reconstruction metrics, Table V reports the radiometric-specific metrics under three color-distortion intensity ranges. DGPIR achieves the best Profile MAE and Profile Corr. across all intensity ranges, indicating stronger cross-track radiometric profile consistency. For NMD, NSD, and SAM, DGPIR remains competitive with the strongest baselines, showing that the proposed method preserves global radiometric statistics while correcting low-frequency residual bias. For example, under the 30–40 intensity range, DGPIR achieves a Profile MAE of 0.0031 and a Profile Corr. of 0.8145, outperforming AdaIR, DFPIR, HAIR, MoCEIR, and PromptIR in cross-track profile preservation.

Fig. 6 presents visual comparisons on representative L1 images affected by low-frequency color distortion. The degraded inputs exhibit large-scale spatially varying color bias. Compared with competing methods, DGPIR more effectively suppresses smooth low-frequency color distortion while preserving intrinsic color information and natural visual consistency.

To further evaluate the adaptability of the global correction branch, we compare DGPIR with traditional column-statistics-based correction methods, including CM, CMA, and CMM. As

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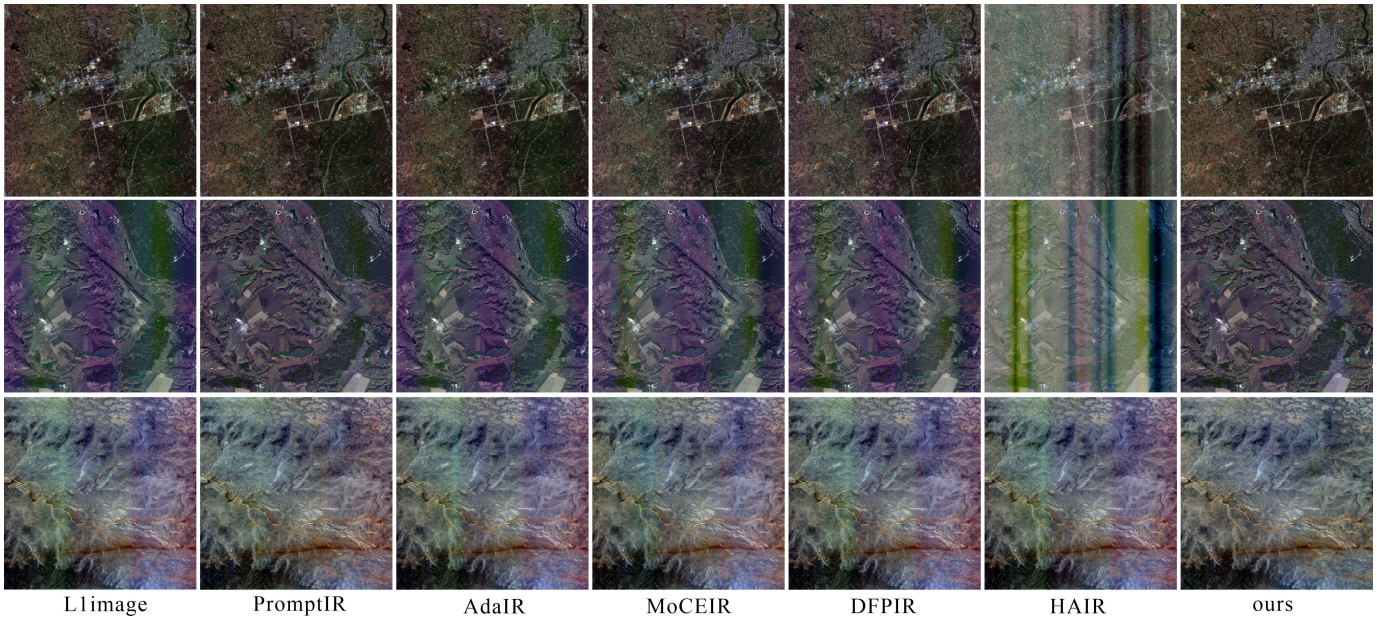


Fig. 6. Visual comparison of low-frequency color distortion correction results produced by different methods on representative L1 remote sensing images. For single-degradation inputs, only the corresponding branch is activated: the global branch is used for low-frequency color distortion, while the patch-based branch is used for striping or noise. For mixed-degradation inputs, DGPIR performs sequential task-conditioned restoration using the same network weights but different task indices. The output of one restoration step is used as the input to the next step following the order from global low-frequency correction to local stripe and noise removal.

TABLE V

RADIOMETRIC-SPECIFIC EVALUATION UNDER LOW-FREQUENCY COLOR DISTORTION. LOWER NMD, NSD, SAM, AND PROFILE MAE INDICATE THE BETTER PERFORMANCE, WHILE HIGHER PROFILE CORR. INDICATES THE BETTER PERFORMANCE

Intensity	Method	NMD ↓	NSD ↓	SAM ↓	Profile MAE ↓	Profile Corr. ↑
10–20	AdaIR	0.0115	0.0226	0.7623	0.0027	0.7556
	DFPIR	0.0106	0.0210	0.8141	0.0025	0.7686
	HAIR	0.0494	0.0599	4.5320	0.0075	0.7667
	MoCEIR	0.0111	0.0219	0.8255	0.0026	0.7896
	PromptIR	0.0107	0.0231	**0.7347**	0.0025	0.8111
	DGPIR	**0.0105**	**0.0204**	0.7642	**0.0023**	**0.8452**
20–30	AdaIR	0.0155	0.0262	0.9442	0.0030	0.7482
	DFPIR	0.0150	0.0247	1.0262	0.0029	0.7499
	HAIR	0.0547	0.0644	5.0551	0.0081	0.7560
	MoCEIR	0.0152	0.0256	1.0384	0.0029	0.7721
	PromptIR	**0.0151**	0.0268	**0.9235**	0.0028	0.8016
	DGPIR	**0.0151**	**0.0242**	0.9806	**0.0026**	**0.8228**
30–40	AdaIR	0.0211	0.0307	1.2041	0.0034	0.7463
	DFPIR	0.0206	0.0296	1.2842	0.0033	0.7391
	HAIR	0.0654	0.0742	5.6327	0.0089	0.7473
	MoCEIR	0.0208	0.0307	1.2964	0.0034	0.7616
	PromptIR	0.0206	0.0314	**1.1867**	0.0032	0.7985
	DGPIR	**0.0205**	**0.0289**	1.2388	**0.0031**	**0.8145**

TABLE VI

COMPARISON WITH TRADITIONAL COLUMN-STATISTICS-BASED CORRECTION METHODS FOR LOW-FREQUENCY COLOR DISTORTION

Method	NMD ↓	NSD ↓	SAM ↓	Profile MAE ↓	Profile Corr. ↑
Degraded	0.0304	0.2159	2.3326	0.0051	0.6316
CM	0.0305	0.2642	2.0403	0.0063	0.4943
CMA	0.0304	0.0689	1.7791	0.0062	0.5329
CMM	0.0305	0.0869	1.7111	0.0063	0.5283
DGPIR	0.0325	0.0482	1.6670	0.0042	0.7600

shown in Table VI, these traditional methods can improve part of the global radiometric statistics, indicating that columnwise statistical normalization is effective for certain low-frequency radiometric deviations.

However, their Profile MAE and Profile Corr. are inferior to those of DGPIR, and in some cases even worse than the degraded input. This suggests that column-statistics-based correction may be affected by scene-dependent radiance variations, especially in land-water mixed scenes or scenes with strong land-cover transitions along the cross-track direction. In comparison, DGPIR achieves better cross-track profile consistency, indicating that the proposed global branch is better able to distinguish low-frequency radiometric residuals from genuine scene radiance variations. Table VII reports the quantitative evaluation under stripe degradations with three intensity regimes: mild, moderate, and severe. Overall, DGPIR delivers consistently strong performance across all regimes, achieving the highest PSNR/SSIM and the lowest RMSE. In the mild stripe regime, DGPIR achieves a PSNR of 62.97 dB, which is 2.09 dB higher than that of the strongest

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TABLE VII
COMPARISON OF RESTORATION PERFORMANCE UNDER STRIPE AT DIFFERENT INTENSITY RANGES (PSNR/SSIM ↑ AND RMSE ↓). BEST RESULTS ARE SHOWN IN BOLD

range	PSNR↑			SSIM↑			RMSE↓		
degraded	51.2433	46.8347	46.1537	0.9925	0.9793	0.9761	133.1456	149.7055	151.6009
promptIR	59.3183	59.2085	57.8583	0.9997	0.9997	0.9997	70.2425	70.2147	80.4919
AdaIR	58.9090	58.7664	58.0125	0.9925	0.9997	0.9997	73.7349	74.4761	81.6948
MoCEIR	59.9271	58.8824	57.6557	0.9998	0.9997	0.9997	66.0943	73.6836	79.6217
DFPIR	60.8883	60.0557	58.5021	0.9998	0.9997	0.9997	58.9962	65.0192	74.5818
HAIR	58.6092	58.3336	57.6036	0.9997	0.9996	0.9996	76.7543	78.3224	80.6845
ours	**62.9713**	**60.2283**	**58.8920**	**0.9999**	**0.9998**	**0.9998**	**47.4136**	**64.0300**	**73.3330**

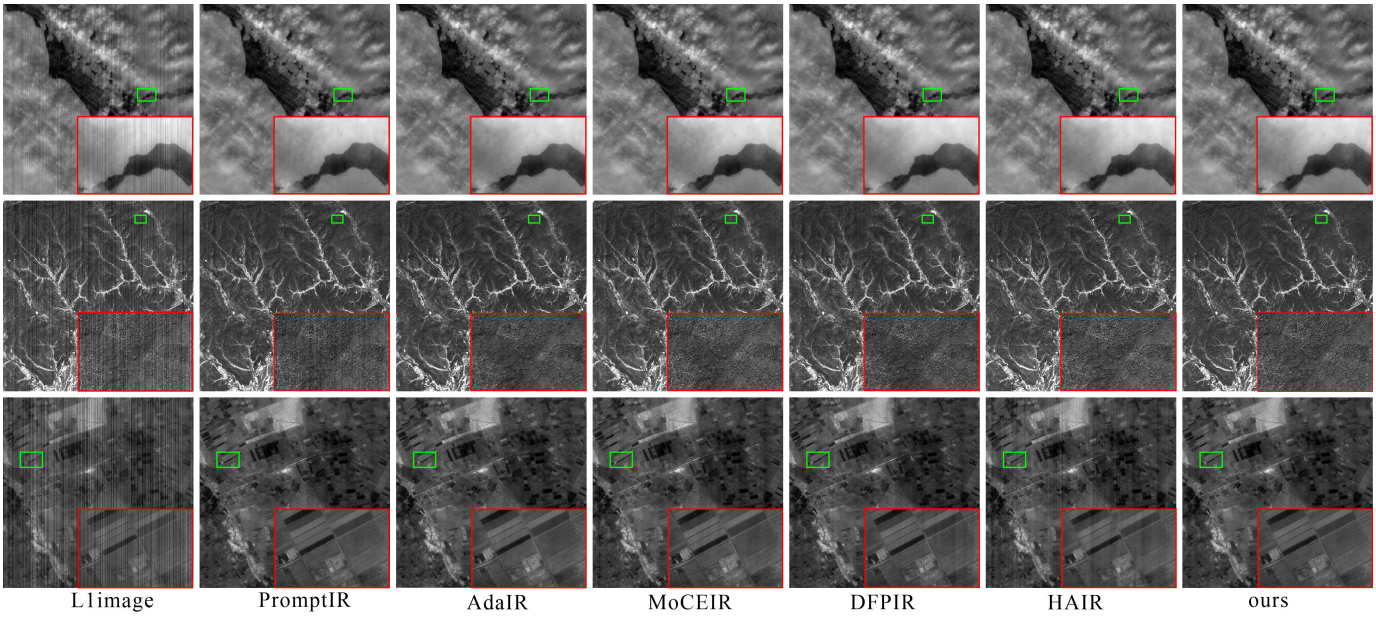


Fig. 7. Visual comparison of destriping results produced by different methods on representative L1 remote sensing images.

TABLE VIII
COMPARISON WITH TASK-SPECIFIC REMOTE SENSING DESTRIPING METHODS ON STRIPE RESIDUAL RESTORATION

Method	PSNR ↑	SSIM ↑	RMSE ↓
TGRS-UTV	52.3108	0.8990	160.0119
TSWEU	52.4027	0.9942	157.2845
WFAF	56.2480	0.9960	107.2414
DGPIR	62.9713	0.9999	47.4136

all-in-one baseline. In addition, RMSE decreases from 58.9962 to 47.4136, indicating improved radiometric reconstruction fidelity under weak stripe interference. As the stripe intensity increases, the performance differences among methods become smaller, which is expected because stronger residual stripes are more difficult to distinguish from inherent scene structures.

Nevertheless, DGPIR remains competitive under severe stripe degradation and still obtains the lowest RMSE. To further compare DGPIR with task-specific destriping algorithms, Table VIII reports the results of three representative remote sensing destriping methods, namely, TGRS-UTV, TSWEU, and WFAF, under the same stripe-residual evaluation setting. Among these task-specific methods, WFAF achieves the strongest performance, with a PSNR of 56.2480, an SSIM of 0.9960, and an RMSE of 107.2414. In comparison, DGPIR achieves a PSNR of 62.9713, an SSIM of 0.9999, and an RMSE of 47.4136, improving PSNR by 6.7233 dB and reducing RMSE by 59.8278 compared with WFAF. These results indicate that although task-specific destriping methods are effective for structured stripe suppression, the proposed task-conditioned framework provides stronger restoration accuracy for weak postcorrection stripe residuals while maintaining compatibility with other L1 residual restoration tasks.

Fig. 7 shows the visual comparison of destriping performance on representative L1 remote sensing scenes covering diverse land-cover types. The enlarged patches show that the

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TABLE IX
COMPARISON OF RESTORATION PERFORMANCE UNDER NOISE AT DIFFERENT INTENSITY RANGES (PSNR/SSIM ↑ AND RMSE ↓). BEST RESULTS ARE SHOWN IN BOLD

range	PSNR			SSIM			RMSE		
degraded	62.2263	56.2801	52.7400	0.9994	0.9975	0.9944	50.7179	96.8604	124.5119
promptIR	56.5485	57.0205	57.1083	0.9993	0.9992	0.9992	92.7169	89.5940	87.3193
AdaIR	60.1443	61.0670	58.9014	0.9996	0.9992	0.9988	64.5028	57.9766	73.8956
MoCEIR	**62.9695**	**62.1441**	**60.3622**	0.9995	**0.9994**	0.9991	**46.6065**	**51.3280**	**62.7008**
DFPIR	61.7708	61.2038	59.1851	0.9996	0.9993	0.9988	53.4093	57.0311	71.4763
HAIR	62.1267	61.4425	59.4759	0.9996	0.9993	0.9989	51.2977	55.5417	69.3092
ours	59.3748	58.3350	58.1958	**0.9997**	0.9993	**0.9993**	74.1774	78.5133	79.6566

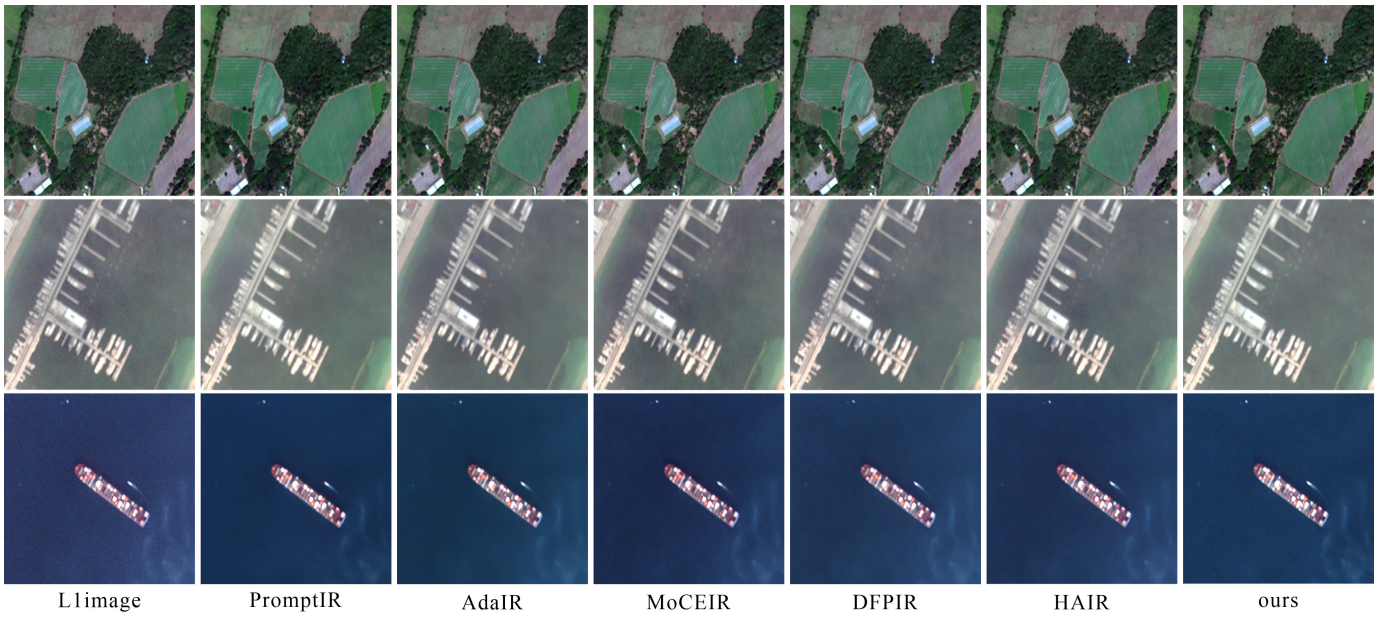


Fig. 8. Visual comparison of denoising results produced by different methods on representative L1 remote sensing images.

input images suffer from vertical stripe residuals with different spatial-frequency characteristics. Compared with competing methods, DGPIR consistently suppresses stripe patterns across test scenes while preserving meaningful scene structures, such as target boundaries and fine-grained textures.

Table IX reports the quantitative denoising results under different noise intensity ranges. DGPIR achieves the highest SSIM in both the low- and high-noise regimes, although its PSNR and RMSE are not always the best among all compared methods. This behavior is reasonable for weak residual noise in L1 imagery, where excessive smoothing may improve pixelwise error but degrade fine structures. DGPIR tends to preserve structural consistency, which is reflected by its higher SSIM and stable performance across noise levels.

Fig. 8 presents visual comparisons of denoising results on representative L1 remote sensing images. DGPIR provides effective noise attenuation while maintaining natural texture appearance across different scene types. This visual observation is consistent with the quantitative results: although DGPIR does not always obtain the best PSNR or RMSE, it maintains competitive pixelwise reconstruction fidelity and favorable structural preservation. In practical L1 imagery, such a tradeoff is desirable because excessive smoothing may reduce discriminative details that are important for visual interpretation and downstream analysis.

D. Evaluation Under Mixed Degradations

In real L1 products, different radiometric residuals may co-exist within the same scene. Such mixed residuals are used here to examine intererror interference. Low-frequency color distortion changes the global and bandwise radiometric baseline, residual striping changes columnwise statistics, and random noise perturbs local estimation cues. These components may therefore affect the estimation and correction of one another. We evaluate DGPIR under three mixed-degradation settings, including low-frequency color distortion plus noise (C + N), striping plus noise (S + N), and the coexistence of low-frequency color distortion, striping, and noise (C + S + N). Table X reports the quantitative results before and after restoration by DGPIR. In these mixed settings, DGPIR performs quality-inspection-guided sequential restoration using the same network weights but different task indices,

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TABLE X
QUANTITATIVE RESULTS OF DGPIR UNDER MIXED-DEGRADATION SETTINGS ON THE KF02B TEST SET. C, S, AND N DENOTE THE LOW-FREQUENCY COLOR DISTORTION, STRIPING, AND RANDOM NOISE, RESPECTIVELY

Mixed setting	Input /Restored	PSNR ↑	SSIM ↑	RMSE ↓	NMD ↓	NSD ↓	SAM ↓	Profile MAE ↓	Profile Corr. ↑
C+N	Degraded input	52.4614	0.9954	156.6845	0.0231	0.0814	2.1686	0.0047	0.6904
C+N	DGPIR	53.2931	0.9983	144.3758	0.0237	0.0344	1.3902	0.0035	0.8120
S+N	Degraded input	52.4736	0.9643	156.0971	0.0005	0.0817	2.0988	0.0045	0.6366
S+N	DGPIR	57.5360	0.9984	89.6253	0.0007	0.0146	0.4572	0.0005	0.9754
C+S+N	Degraded input	52.1050	0.9640	162.8237	0.0249	0.1280	2.9589	0.0066	0.5542
C+S+N	DGPIR	53.0168	0.9980	148.2453	0.0251	0.0487	1.9147	0.0043	0.7374

following the order from global low-frequency correction to local stripe and noise removal.

The results show that DGPIR consistently improves restoration quality under all three mixed settings. For C + N, PSNR increases from 52.4614 to 53.2931 dB, SSIM increases from 0.9954 to 0.9983, and RMSE decreases from 156.6845 to 144.3758. For S + N, PSNR increases from 52.4736 to 57.5360 dB, SSIM increases from 0.9643 to 0.9984, and RMSE decreases from 156.0971 to 89.6253. Under the more challenging C + S + N setting, DGPIR also improves PSNR from 52.1050 to 53.0168 dB and SSIM from 0.9640 to 0.9980, while reducing RMSE from 162.8237 to 148.2453.

In addition to improving pixelwise fidelity, DGPIR also improves radiometric consistency under mixed degradations. For example, under S + N, SAM decreases from 2.0988 to 0.4572, Profile MAE decreases from 0.0045 to 0.0005, and Profile Corr. increases from 0.6366 to 0.9754. Under the more challenging C + S + N setting, SAM decreases from 2.9589 to 1.9147, and Profile Corr. increases from 0.5542 to 0.7374. These results indicate that DGPIR improves not only pixelwise reconstruction quality but also cross-track radiometric profile consistency when multiple residual categories coexist. Under mixed residuals, the correction of one residual component can influence the statistical cues used by subsequent restoration steps. The global-to-local restoration order first estimates the low-frequency radiometric baseline and then suppresses column-dependent stripes and local noise, so that stripe and noise restoration are performed on images with reduced large-scale radiometric bias. Since each pass is conditioned on one quality-inspection-derived residual category, DGPIR avoids compressing heterogeneous residuals into a single undifferentiated prompt while still reusing a shared radiometric representation. The smaller improvement under C + S + N compared with S + N indicates that the proposed strategy alleviates, but does not completely eliminate, degradation coupling and error accumulation between sequential stages.

E. Performance and Robustness Analysis

To further assess the stability of the proposed model across different degradation types and severity levels, Fig. 9 presents the per-sample distributions of PSNR and RMSE before and after restoration for three representative tasks, namely, low-frequency color distortion correction, destriping, and denoising. Different colors denote different degradation ranges within each task.

In Fig. 9(a)–(f), the scatter plots compare the postrestoration PSNR against the prerestoration PSNR, and the corresponding histograms below summarize the distribution of. For low-frequency color distortion correction and destriping, most samples are distributed above the diagonal line, indicating that the restored images achieve higher PSNR than the degraded inputs. Consistently, the histograms are mainly concentrated in the positive range, confirming stable performance gains across different degradation levels. Among the three tasks, destriping exhibits the most pronounced positive shift, suggesting strong stripe-suppression capability. For denoising, substantial PSNR improvements are observed for medium- and high-noise samples, whereas the low-noise group shows a distribution closer to zero, with a subset of samples exhibiting slightly negative gains. This reflects a common tradeoff in restoration models, where aggressive noise suppression under very mild degradation may slightly attenuate fine textures. Fig. 9(g)–(l) shows the corresponding RMSE analysis, where the improvement is defined such that positive values indicate error reduction. Most samples lie below the diagonal line in the RMSE scatter plots, demonstrating that the RMSE after restoration is generally lower than that of the degraded input. The histograms are likewise biased toward positive values across all three tasks, further confirming that the proposed method consistently reduces reconstruction error under different degradation conditions. Notably, the positive shifts remain observable across different degradation ranges within each task, indicating that the proposed framework maintains stable restoration behavior rather than benefiting only from a specific degradation level. These results suggest that DGPIR is robust to heterogeneous L1 radiometric residual categories and degradation strengths.

F. Cross-Orbit and Cross-Sensor Physical Consistency Analysis

For push-broom optical satellite sensors, residual low-frequency color distortion within the same orbit is generally expected to exhibit certain consistency, since it is often

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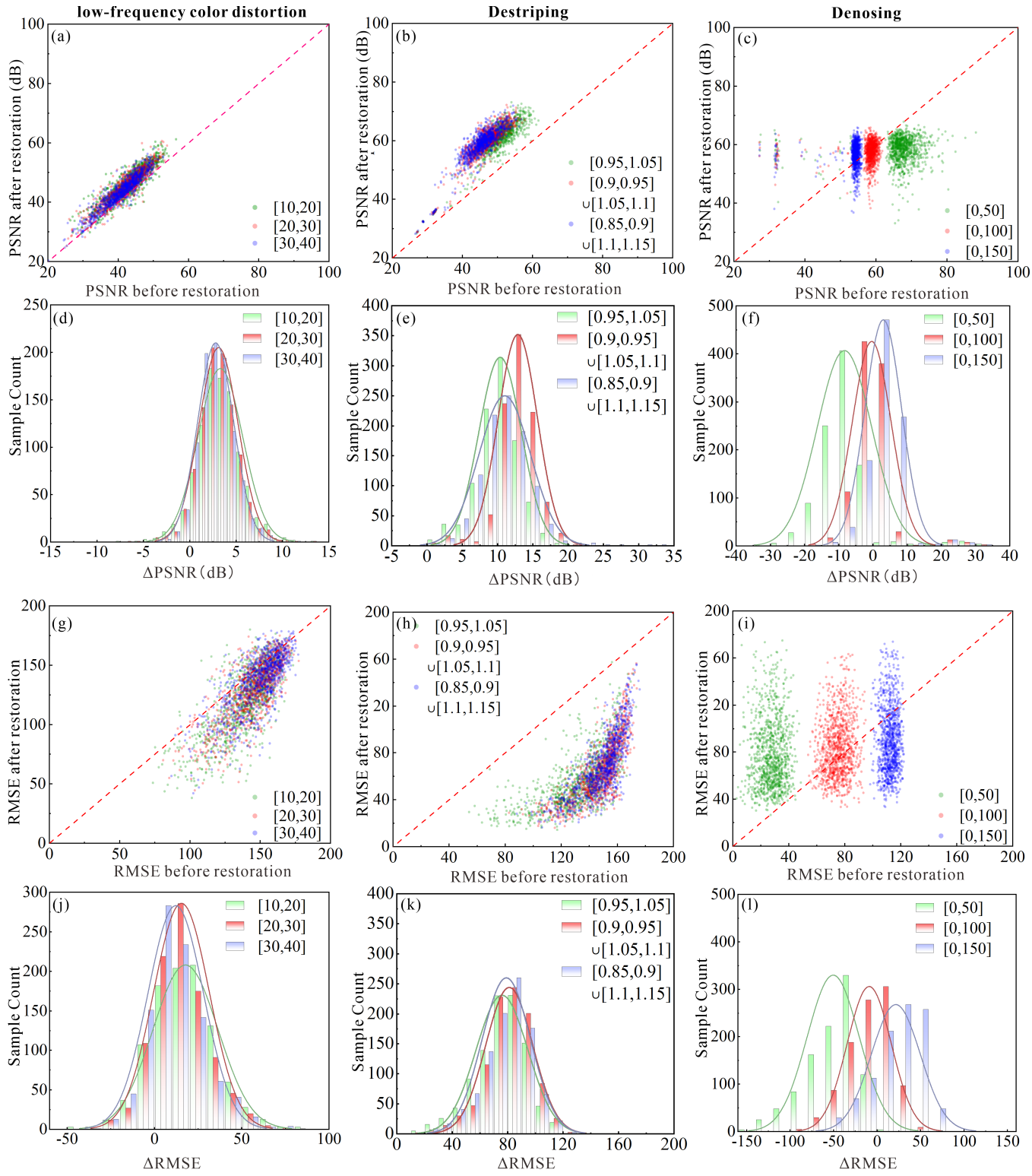
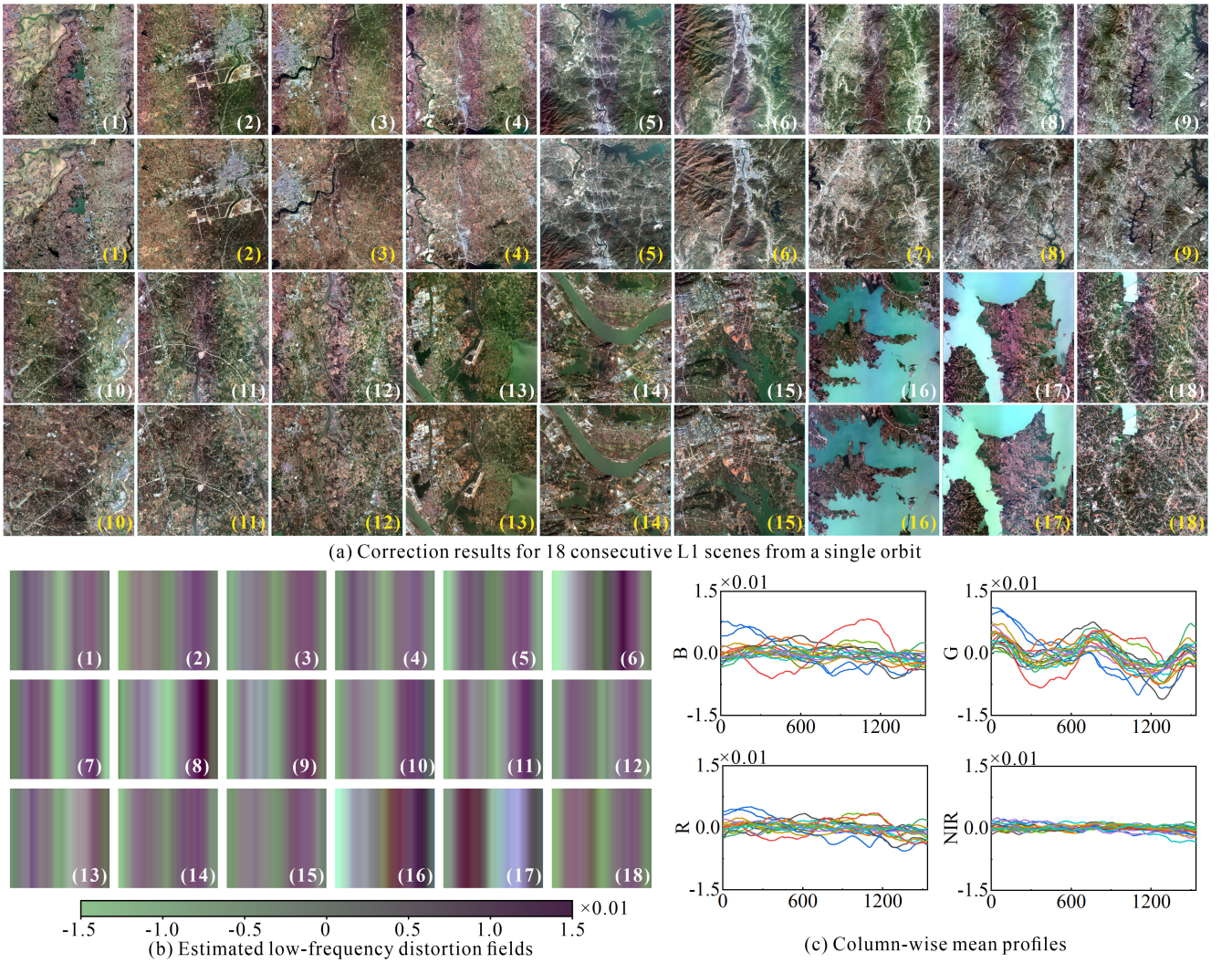


Fig. 9. Samplewise performance distributions of the three restoration tasks on the test set. (a), (d), (g), and (j) Low-frequency color distortion correction, (b), (e), (h), and (k) destriping, and (c), (f), (i), and (l) denoising. In each scatter plot, the x-axis denotes the metric before restoration, and the y-axis denotes the metric after restoration; the dashed diagonal indicates equal performance before and after restoration. The histograms show the distributions of ΔPSNR (dB) and ΔRMSE, where positive ΔPSNR and ΔRMSE reduction indicate performance improvement.

associated with slowly varying sensor-related calibration residuals. To examine whether DGPIR can capture such physically consistent residual patterns, we apply the model to a continuous sequence of 18 L1 scenes acquired along a single orbit. As shown in Fig. 10(a), these scenes cover diverse land-cover types, including urban areas, mountainous vegetation, and open water bodies, while exhibiting visually similar low-frequency color distortions. Fig. 10(b) presents the

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(a) Correction results for 18 consecutive L1 scenes from a single orbit

(b) Estimated low-frequency distortion fields

(c) Column-wise mean profiles

Fig. 10. Orbit-level consistency analysis on 18 consecutive L1 scenes from a single orbit. (a) Restoration results for 18 scenes from the same orbit. (b) Estimated low-frequency correction fields produced by DGPIR, where the color scale denotes normalized gain bias. (c) Columnwise mean profiles of the estimated correction fields in the B, G, R, and NIR bands. The clustered profiles indicate that DGPIR captures a stable cross-track residual pattern shared by scenes from the same orbit.

corresponding low-frequency distortion fields estimated by DGPIR. Despite substantial semantic variation among the scenes, the estimated distortion fields retain similar columnwise gradient patterns, suggesting that the model can separate scene-dependent image content from sensor-related low-frequency radiometric residuals.

To further verify this orbit-level consistency, we compute the columnwise mean profiles of the estimated distortion fields for the B, G, R, and NIR bands, as shown in Fig. 10(c). The profiles extracted from different scenes are tightly clustered and exhibit similar shapes and amplitudes across the cross-track direction. This observation indicates that DGPIR captures a stable radiometric residual pattern shared by scenes from the same orbit, rather than simply fitting scene-specific textures or land-cover structures.

Based on the additional validation data summarized in Table III, we further analyze the cross-sensor applicability of DGPIR using the estimated low-frequency gain profiles. For each orbit group, the predicted low-frequency correction field is averaged along the along-track direction to obtain 1-D cross-track gain profiles for the RGB bands. The solid curves in Fig. 11 denote the mean estimated gain profiles, and the shaded bands denote the mean $\pm$ one standard deviation across scenes within the same orbit group at each column position.

The results show that low-frequency color distortion does not necessarily appear in all spectral bands. In several cases, only one or two bands exhibit obvious low-frequency gain variation, which is sufficient to produce visible color distortion in the restored RGB composite. The affected bands show larger profile amplitudes and stronger consistency across scenes, whereas bands without obvious color distortion present weaker amplitudes and more random fluctuations. This band-dependent behavior is consistent with residual interband or interchannel response differences in practical L1 products.

Fig. 11(a) and (b) corresponds to two KF02B05 orbit groups acquired by PMS cameras. In (a), the green band shows the most evident low-frequency gain variation, while the blue and red bands remain relatively weak. In (b), both the green

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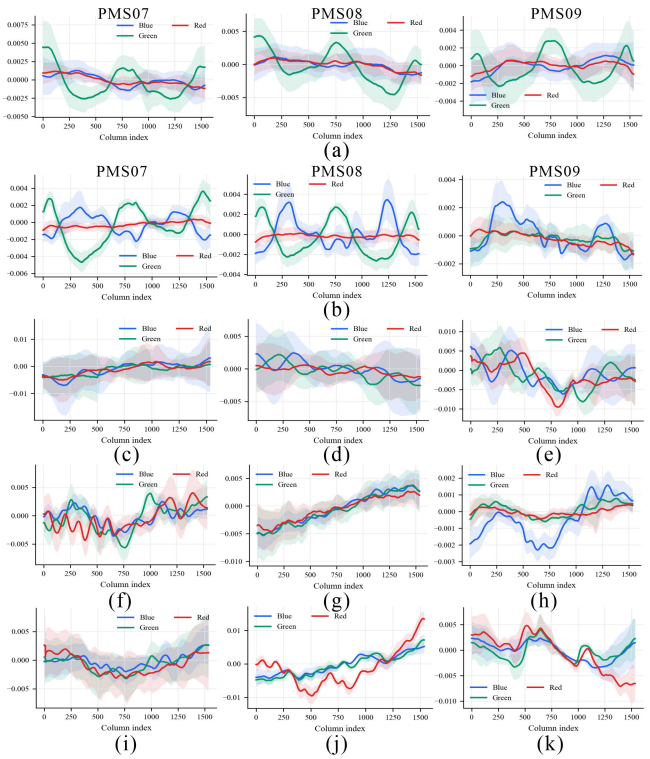


Fig. 11. Estimated low-frequency gain profiles on additional real L1 validation scenes from different orbit groups and satellite platforms. The profiles are obtained by averaging the predicted low-frequency correction fields along the along-track direction. In each subpanel, the solid blue, green, and red curves denote the mean cross-track gain profiles of the corresponding RGB bands over all scenes within the same orbit group. The shaded bands denote the inter-scene variation band, defined as the mean $\pm$ one standard deviation at each column position. (a) and (b) KF02B05 orbit groups acquired by PMS cameras. (c)–(k) GF03D-series scenes. Bands with evident low-frequency residuals show larger amplitudes and stronger cross-scene consistency, whereas weakly affected bands show lower amplitudes and more random fluctuations. (c)–(e) Limited-correction cases, suggesting that transfer performance depends on whether the real residual profile falls within the learned degradation space.

and blue bands exhibit noticeable low-frequency variations. Fig. 11(c)–(k) correspond to GF03D-series scenes acquired over different regions and imaging conditions. Although the profile shapes vary across these cases, DGPIR can estimate structured low-frequency correction profiles for several GF03D scenes, as shown in (f)–(k).

The cases shown in Fig. 11(c)–(e) exhibit limited correction quality. Their estimated profiles have weaker repeatability, larger irregular fluctuations, or amplitudes that deviate from the dominant patterns observed in the training degradation space. These observations indicate that DGPIR has practical transferability when the target sensor residuals follow structured low-frequency cross-track response patterns similar to those represented in the simulated degradation space. In contrast, out-of-distribution residual forms may require additional sensor-specific samples, calibration knowledge, or an expanded degradation simulation range. Therefore, the applicability of DGPIR to other satellite platforms depends on the consistency between the target residual morphology and the degradation space covered during training, rather than on the satellite name or sensor platform alone.

TABLE XI

QUANTITATIVE CONSISTENCY METRICS OF ESTIMATED LOW-FREQUENCY GAIN PROFILES ON ADDITIONAL REAL L1 ORBIT GROUPS

	mean_intra_band_corr				pca_evr1_zscore			
	b1	b2	b3	b4	b1	b2	b3	b4
1	0.86	0.51	0.26	0.10	0.88	0.59	0.34	0.31
	0.12	0.58	0.18	0.06	0.35	0.63	0.39	0.37
	0.16	0.43	0.08	0.14	0.39	0.50	0.35	0.32
2	0.68	0.90	0.18	0.10	0.73	0.91	0.41	0.44
	0.63	0.80	0.21	0.06	0.67	0.82	0.45	0.28
	0.60	0.12	0.28	0.26	0.66	0.37	0.47	0.46
3	0.07	0.04	0.07	0.08	0.82	0.96	0.76	0.57
4	0.12	0.10	0.03	0.01	0.53	0.54	0.57	0.36
5	0.12	0.22	0.22	0.39	0.69	0.62	0.62	0.59
6	0.54	0.50	0.27	0.38	0.60	0.58	0.50	0.51
7	0.74	0.35	0.26	(0.00)	0.79	0.61	0.76	0.45
8	0.86	0.51	0.26	0.10	0.88	0.59	0.34	0.31
9	0.31	0.19	0.15	0.09	0.50	0.42	0.33	0.35
10	0.68	0.88	0.94	0.79	0.78	0.90	0.95	0.84
11	0.54	0.50	0.27	0.38	0.60	0.58	0.50	0.51
12	0.98	0.79	0.61	0.40	0.99	0.83	0.69	0.61

Table XI reports the quantitative consistency of the estimated low-frequency gain profiles on additional real L1 orbit groups. The quantitative results are consistent with the profile visualization in Fig. 11. For the KF02B05 orbit groups, the dominant affected bands show clearly higher profile consistency. For example, in one PMS07 group, the B-band Profile Corr. reaches 0.857, with a PCA-EVR1 of 0.883, indicating a stable and repeatable low-frequency mode. In another KF02B05 group, the G-band Profile Corr. reaches 0.901 for PMS07 and 0.800 for PMS08, with corresponding PCA-EVR1 values of 0.910 and 0.820, respectively. These results are consistent with the visual observation that different PMS cameras or orbit groups may exhibit different dominant affected bands.

For the GF03D-series scenes, the consistency varies across acquisition conditions. Several groups still show evident low-dimensional profile structure, as reflected by relatively high PCA-EVR1 values even when pairwise Profile Corr. are moderate. This suggests that the model can capture structured cross-track low-frequency patterns on satellite platforms not used in the original KF02B training set. Meanwhile, groups with low Profile Corr. or unstable profile amplitudes correspond to more challenging cases, where the estimated residual fields are less repeatable across scenes. These results indicate that DGPIR has cross-sensor applicability when the target residual follows a structured low-frequency cross-track response pattern, while its transfer performance is limited for residuals with abnormal amplitudes or irregular profile shapes.

To further examine whether the simulated degradation space is consistent with real L1 residual patterns, we compare the gain profiles estimated from real L1 scenes with gain profiles

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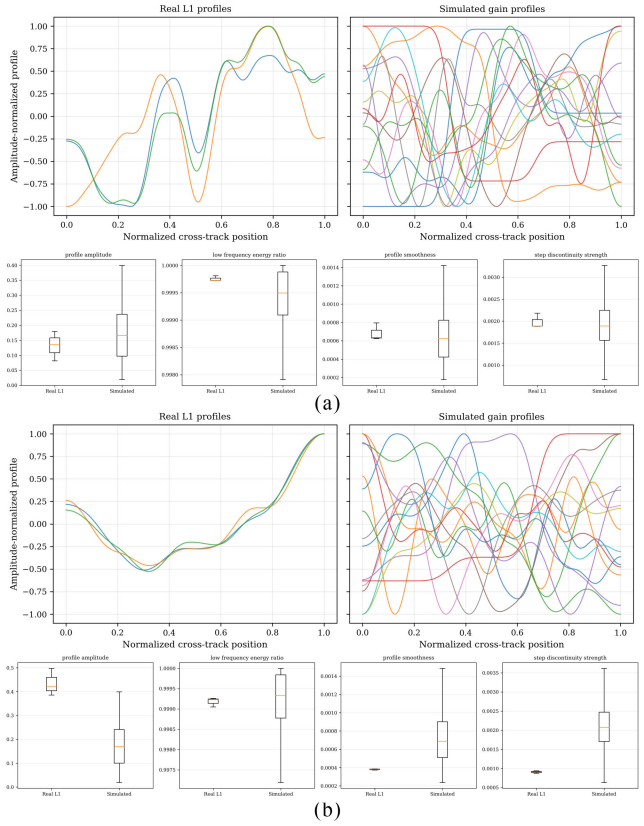


Fig. 12. Comparison between estimated real L1 gain profiles and simulated gain profiles. (a) Stable-correction case. (b) Limited-correction case. (Top) Amplitude-normalized cross-track gain profiles from real L1 scenes and simulated degradation samples, where normalization is used only for shape comparison. (Bottom) Comparison of four profile-level descriptors. Profile amplitude is calculated from the original gain profile as the 5th-to-95th percentile range normalized by the absolute mean profile value. The low-frequency energy ratio is computed from the mean-centered profile as the proportion of non-DC FFT energy below the low-frequency cutoff to the total non-DC profile energy. Profile smoothness is measured by the mean absolute first-order difference of the normalized profile, and step-discontinuity strength is measured by the 95th percentile of the absolute first-order difference. Stable-correction cases show descriptor distributions close to the simulated degradation space, whereas limited-correction cases deviate in amplitude or profile shape.

generated by the proposed degradation simulation model. The comparison is conducted at the profile-statistic level rather than at the image-pixel level, because low-frequency radiometric residuals are mainly characterized by their cross-track gain variation. For each profile, we compute four descriptors: profile amplitude, low-frequency energy ratio, profile smoothness, and step-discontinuity strength. These descriptors jointly characterize the magnitude, frequency distribution, smoothness, and local discontinuity of the gain field.

As shown in Fig. 12, the real L1 profiles with stable correction are generally close to the simulated gain-profile distribution in terms of amplitude and low-frequency structure.

Their profiles mainly exhibit smooth cross-track variation or moderate step-like transitions, which are consistent with the two components used in the degradation model. In contrast, the limited-correction cases, corresponding to Fig. 11(c)–(e), show profile statistics that deviate from the simulated distribution. In particular, their amplitudes, smoothness, or discontinuity strength are outside the dominant range of the simulated samples.

This observation provides an empirical link between the physical degradation model and the learned restoration behavior. When the real residual gain profile falls within the simulated degradation space, DGPIR can estimate stable low-frequency correction fields. When the residual profile deviates from the simulated distribution, the correction becomes less reliable. Therefore, the proposed degradation simulation is physically relevant to the dominant low-frequency radiometric residuals observed in representative L1 products, while the remaining out-of-distribution cases indicate the direction for further expanding the degradation model.

G. Ablation Studies

To evaluate the contribution of the key architectural components, we conduct ablation experiments on the KF02B synthetic test set, as summarized in Table XII. The evaluated components include the task-adaptive prompt, PAA, PGF, and the stagewise prompt sharing strategy. In addition to the baseline model, we introduce an ablation variant denoted as Ours-no_stage_share, in which prompts are estimated independently for each Transformer layer instead of being shared within each stage.

Compared with the baseline model, introducing the task-adaptive prompt improves low-frequency color distortion correction from 52.15 to 53.43 dB and substantially improves radiometric consistency metrics, reducing NMD from 0.052 to 0.022 and Profile MAE from 0.0055 to 0.0034. This indicates that explicitly incorporating residual-category information helps the network adjust its restoration behavior according to the target residual category. Meanwhile, the stripe and noise PSNR values do not improve uniformly in this variant, suggesting that prompt guidance alone is not sufficient to achieve the best balance across all residual categories.

Based on the prompt-guided framework, we further investigate the effects of PAA and PGF. With PAA, the model achieves the best low-frequency PSNR among the ablation variants, increasing from 53.43 to 53.81 dB, and also improves stripe restoration from 58.23 to 59.79 dB. This suggests that PAA is beneficial for modeling spatially structured residual patterns. With PGF, the model achieves competitive low-frequency correction and stripe restoration while improving several radiometric consistency metrics, indicating that prompt-gated nonlinear feature transformation contributes to degradation-adaptive restoration.

Finally, by jointly integrating the task-adaptive prompt, PAA, PGF, and stagewise prompt sharing, the full model achieves the best balance between stripe restoration and radiometric profile consistency. Compared with Ours-no_stage_share, the full model improves stripe PSNR from 58.47 to 60.08 dB, increases Profile Corr. from 0.788 to 0.814, and reduces Profile MAE from 0.0033 to 0.0031. Since the two variants use the same network parameters and have nearly identical inference cost, the improvement mainly comes from more consistent prompt guidance rather than increased model capacity. Although Ours-no_stage_share obtains a higher

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TABLE XII

ABLATION STUDY OF THE PROPOSED DGPIR ON THREE RESTORATION TASKS AND RADIOMETRIC CONSISTENCY METRICS. BEST RESULTS ARE SHOWN IN BOLD

Model	Prompt	PAA	PGF	Low PSNR	Stripe PSNR	Noise PSNR	Low SSIM	Stripe SSIM	Noise SSIM	NMD ↓	NSD ↓	SAM ↓	Profile MAE ↓	Profile Corr. ↑
baseline	×	×	×	52.15	**60.77**	**60.88**	0.9966	**0.9995**	**0.9992**	0.052	0.035	1.74	0.0055	0.739
M1/only prompt	✓	×	×	53.43	58.23	60.14	0.9985	0.9991	0.9990	0.022	0.031	1.19	0.0034	0.778
M2/attn	✓	✓	×	53.81	59.79	57.62	0.9986	0.9993	0.9989	**0.021**	**0.029**	**1.23**	0.0032	0.805
M3/gdfn	✓	×	✓	53.67	59.60	56.98	0.9986	0.9993	0.9988	0.020	0.030	1.20	0.0032	0.789
Ours/no stage share	✓	✓	✓	53.63	58.47	59.63	0.9986	0.9992	0.9988	0.020	**0.029**	**1.23**	0.0033	0.788
Ours	✓	✓	✓	**53.79**	60.08	57.48	**0.9987**	0.9994	0.9988	**0.021**	**0.029**	1.24	**0.0031**	**0.814**

TABLE XIII

SENSITIVITY ANALYSIS OF BRANCH B INFERENCE RESOLUTION FOR LOW-FREQUENCY COLOR DISTORTION CORRECTION. NMD, NSD, SAM, AND PROFILE MAE ARE LOWER BETTER, WHILE PROFILE CORR. IS HIGHER BETTER

Resolution	NMD ↓	NSD ↓	SAM ↓	Profile MAE ↓	Profile Corr. ↑
512	**0.0236**	**0.0312**	1.3477	**0.0034**	0.8121
768	0.0237	0.0315	1.3539	0.0034	**0.8132**
1024	0.0238	0.0317	1.3580	0.0035	0.8117
1536	0.0237	0.0319	1.3580	0.0035	0.8069
2048	0.0236	0.0320	1.3515	0.0035	0.8008
2560	0.0237	0.0321	**1.3376**	0.0035	0.7968

PSNR on the denoising subset, its lower stripe-removal performance and weaker radiometric profile consistency indicate that independently assigned layerwise prompts may provide less stable degradation guidance across network depth. Therefore, stagewise prompt sharing is adopted in the final model to improve restoration stability and radiometric consistency.

We further conduct a sensitivity analysis on the fixed inference resolution of Branch B for low-frequency color distortion correction. Since low-frequency color distortion spans the full image swath, directly replacing Branch B with a patch-based inference path would discard global cross-track radiometric context and may cause inconsistent correction between patches. Therefore, we evaluate different global inference resolutions from 512 to 2560 pixels to examine the tradeoff between radiometric accuracy and computational cost.

As shown in Table XIII, increasing the fixed inference resolution does not lead to consistent improvement. The 512-pixel setting achieves the lowest NMD, NSD, and Profile MAE, while the 768-pixel setting obtains the highest Profile Corr. The differences among 512, 768, and 1024 are marginal, indicating that the low-frequency radiometric field can be reliably estimated at a moderate downsampled resolution. Considering that the 512-pixel setting provides competitive or best radiometric consistency with lower memory consumption, it is adopted as the default resolution for Branch B.

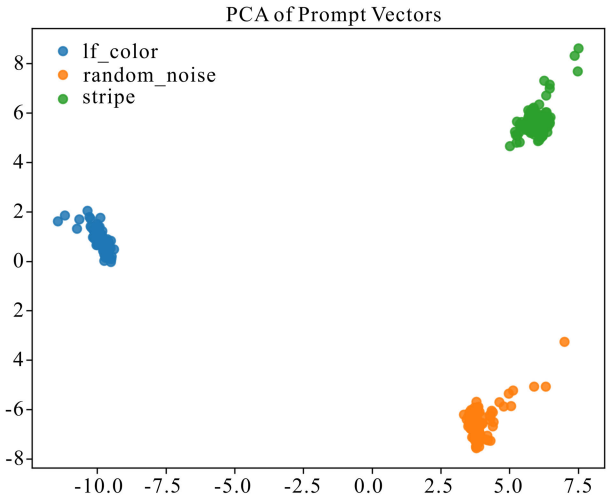


Fig. 13. PCA visualization of prompt vectors generated by DGGP for low-frequency color distortion, striping, and random noise.

V. DISCUSSION

A. Mechanistic Interpretation of DGPIR

Although all-in-one restoration frameworks such as PromptIR have shown strong performance on natural-image benchmarks, their direct application to L1 remote sensing products remains challenging because the underlying residuals are physically heterogeneous. Radiometric residuals in optical satellite imagery differ substantially in spatial distribution, frequency characteristics, and radiometric manifestation. As a result, optimizing a shared network without explicit task conditioning may lead to feature interference, where improving one restoration objective can compromise the feature representations required by another. This issue is particularly relevant for L1 restoration, where low-frequency color correction, stripe suppression, and denoising impose different constraints on the learned features.

The proposed DGPIR framework addresses this issue by converting the under-constrained multidegradation restoration problem into a quality-inspection-guided degradation-aware restoration process. The ablation results show that explicit

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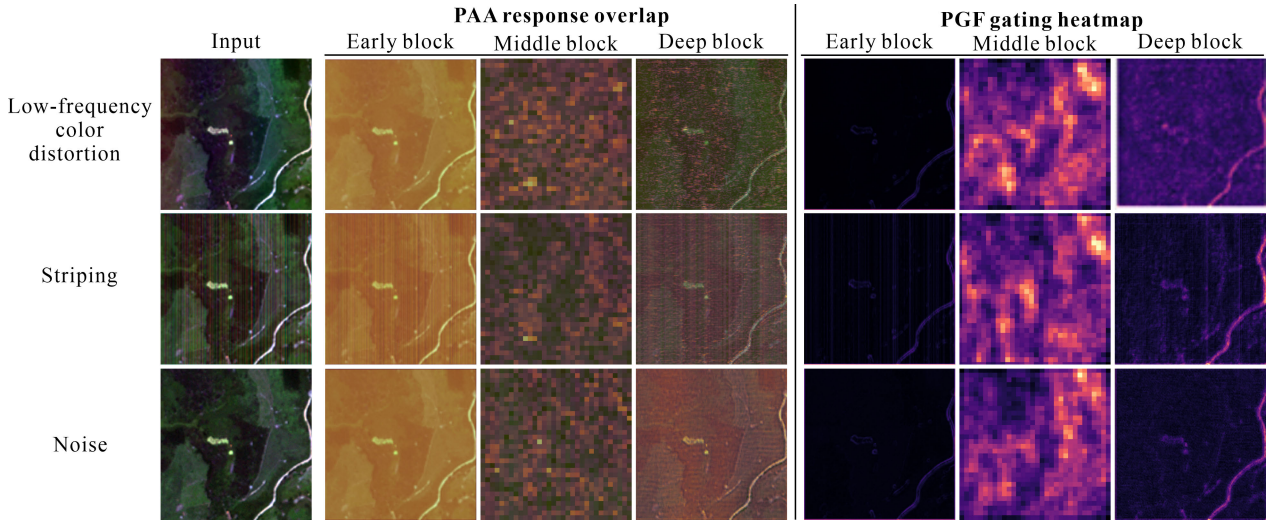


Fig. 14. Visualization of prompt-induced modulation in DGPIR. PAA attention-response maps and PGF gating heatmaps are shown for low-frequency color distortion, striping, and random noise across early, middle, and deep blocks. The category-dependent responses indicate that prompts guide the backbone toward degradation-specific feature modulation.

degradation guidance is important for reducing task interference in the unified restoration framework. Rather than relying on a single implicit feature space to accommodate all residual modes, the prompt-guided design encourages the model to adapt its restoration behavior according to the identified radiometric residual category and image-derived degradation statistics.

To further interpret this behavior, we visualize the prompt consistency between the target residual morphology and the degradation space covered during training, rather than on the satellite name or sensor platform alone distributions and prompt-induced modulation responses, as shown in Figs. 13 and 14. The prompt vectors generated for low-frequency color distortion, striping, and random noise form distinguishable distributions in the embedding space, indicating that DGGP learns degradation-dependent prompt representations rather than collapsing different residual categories into a single undifferentiated guidance signal. The PAA attention-response maps show category-dependent activation patterns, while the PGF gating heatmaps reveal task-dependent feature selection across different network depths.

These observations provide intuitive evidence that the prompts guide the backbone to adjust its attention and nonlinear feature transformation according to the corresponding residual category.

A further insight is that the network architecture should be aligned with the physical properties of the target residuals. Residual stripes in push-broom imagery exhibit strong directional continuity and long-range spatial correlation, making them well suited to the enhanced dependency modeling introduced by the PAA module. By contrast, low-frequency color distortion is better characterized as a slowly varying radiometric bias, for which prompt-gated feature transformation can provide adaptive channelwise modulation. These complementary components suggest that the gains of DGPIR do not arise simply from increased model complexity, but from incorporating architectural priors that better match the physical structure of different residual categories.

The mixed-degradation results further suggest that DGPIR does not exhibit severe prompt conflict when multiple radiometric residual categories are present. This is mainly because each restoration pass is conditioned on one quality-inspection-derived residual category, while mixed cases are handled through sequential restoration from global to local residuals. However, the relatively smaller improvement under C + S + N compared with S + N indicates that compound degradation remains more difficult than single or two-factor degradation, especially when low-frequency radiometric bias, structured stripes, and random noise are coupled in the same image.

The orbit-level and cross-sensor profile analyses further examine whether the estimated correction fields remain physically consistent beyond paired synthetic tests. The stable cross-track profiles observed in several orbit groups indicate that DGPIR can capture scene-independent low-frequency residual patterns, whereas unstable profiles identify cases outside the current learned degradation space.

The denoising results also reveal a tradeoff between restoration strength and detail preservation under extremely mild degradation, where aggressive noise suppression may slightly attenuate fine textures.

B. *Efficiency Analysis*

Table XIV compares the computational efficiency of DGPIR with representative all-in-one restoration methods in terms of parameter count, FLOPs, peak GPU memory usage, and inference time. DGPIR has the largest parameter count among the compared methods, with 37.16 M parameters. This mainly reflects the use of a high-capacity Transformer backbone together with task-conditioned prompt modulation modules, which are introduced to support unified restoration across

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TABLE XIV
COMPARISON OF MODEL COMPLEXITY AND INFERENCE EFFICIENCY AMONG DIFFERENT ALL-IN-ONE RESTORATION METHODS

Method	Params (M)	FLOPs (G)	GPU memory(MB)	Inference Times(ms)
PromptIR	32.964	157.971	710.81	94.97
AdaIR	28.720	147.019	676.47	104.96
MoCEIR	23.523	74.712	342.67	66.75
DFPIR	26.094	140.870	667.04	87.65
HAIR	7.640	41.061	676.47	102.32
ours	37.161	95.842	424.18	97.50

low-frequency color distortion, residual striping, and weak random noise.

Despite its relatively large parameter count, DGPIR maintains a moderate computational cost. Its FLOPs are 95.84 G, which is substantially lower than those of PromptIR, AdaIR, and DFPIR, while higher than the lighter MoCEIR and HAIR models. This indicates that the proposed architecture does not achieve improved restoration performance simply by increasing computation, but instead benefits from degradation-guided feature modulation within a shared restoration framework.

A similar trend can be observed in peak GPU memory usage. DGPIR requires 424.18 MB during inference, which is considerably lower than PromptIR, AdaIR, DFPIR, and HAIR, although slightly higher than MoCEIR. In terms of inference time, DGPIR is not the fastest model, but its runtime of 97.5 ms remains within a practical range and is comparable to PromptIR while being faster than AdaIR and HAIR. Overall, these results show that DGPIR provides a favorable tradeoff between restoration capability and computational efficiency, making it suitable for practical L1 radiometric restoration under large-swath inference scenarios.

C. Limitations

Several limitations remain and define important directions for future work. First, distinguishing authentic high-frequency scene textures from weak structured residuals remains challenging. In complex scenes containing dense urban patterns, agricultural textures, or fine linear structures, weak residual stripes may share similar directional or frequency characteristics with real land-cover details. Although the proposed prompt-guided framework improves degradation-aware restoration, it still mainly relies on spectral–spatial feature responses and does not explicitly incorporate high-level semantic or content-aware priors. As a result, a small number of cases may still suffer from slight over-smoothing when suppressing structured residuals. Future work may introduce semantic guidance or content-adaptive regularization to better preserve texture-rich regions during restoration.

Second, the current framework assumes that the quality-inspection-derived residual category is consistent with the dominant degradation pattern in the input image. This assumption is reasonable for practical L1 production workflows, where scenes are usually inspected by trained operators or automated quality-control algorithms before restoration. However, inaccurate residual-category classification, ambiguous residual categories, or previously unseen residual patterns may lead to suboptimal prompt selection. Extending DGPIR toward more flexible blind or semi-blind restoration would require automatic degradation perception, confidence-aware task selection, or a mixture-of-prompts mechanism to adaptively handle uncertain degradation conditions.

Third, the degradation simulation space remains limited. The current synthetic models cover representative low-frequency, stripe, and noise patterns in wide-swath push-broom imagery, but real L1 products may contain sensor-specific response changes, orbit-dependent illumination effects, or 2-D residual fields outside this space. Such out-of-distribution profiles may reduce transfer reliability; expanding the degradation space with sensor-specific samples, calibration knowledge, and real quality-inspection feedback is therefore necessary.

VI. CONCLUSION

This article presented DGPIR, a degradation-guided all-in-one framework for postcorrection radiometric restoration of L1 optical satellite imagery. The proposed method uses quality-inspection-derived radiometric residual categories and image-derived degradation statistics to generate dynamic scale-specific prompts, which modulate a shared Transformer backbone through PAA and PGF modules. In addition, a dual-branch inference strategy is designed to match the physical scale of different residuals, where the global branch corrects full-swath low-frequency color distortion, and the patch-based branch handles residual striping and weak random noise.

Experiments on representative L1 restoration tasks show that DGPIR achieves competitive or superior performance compared with state-of-the-art all-in-one restoration baselines, with clear advantages in low-frequency correction, destriping, radiometric consistency, and structural preservation. The results show that DGPIR improves low-frequency color distortion correction, provides effective stripe suppression, maintains stable denoising behavior, and remains effective under mixed-degradation settings. The ablation studies further verify the contributions of degradation-guided prompts, PAA, PGF modulation, and stagewise prompt sharing. Moreover, the cross-orbit and cross-sensor analyses indicate that the estimated low-frequency correction fields exhibit physically meaningful cross-track consistency, suggesting the applicability of the proposed framework to practical L1 radiometric quality refinement.

Future work will focus on uncertain residual categories, unseen residual patterns, more complex mixed degradations, and broader cross-sensor validation.

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